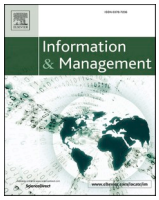
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# Toward multilevel research on private information disclosure: A systematic review and future agenda

Mingxin Zhang *, Reeva Lederman, Ofir Turel

*The University of Melbourne, Parkville, VIC 3052, Australia*

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ABSTRACT

Through a systematic literature review, we unravel current theorizing on private information disclosure (PID). We note that existing models of PID have mainly taken an individual-level perspective. In line with current understanding of privacy management, we posit that PID should be more systematically understood as a multilevel phenomenon, for two reasons. First, the disclosed information sometimes concerns multiple stakeholders beyond oneself. Second, disclosing private information can often be affected by hierarchically structured (nested) humans within social units. To this end, we propose a multilevel theoretical framework together with future recommendations, aimed at inspiring future multilevel PID research.

## 1. Introduction

Private information disclosure (PID) captures the sharing of sensitive personal information that one does not usually reveal to the public [1–4]. It often occurs on social networking sites, digital health platforms, and e-commerce websites, where individuals share sensitive personal information with online peers ("horizontal" information sharing) or online service providers ("vertical" information sharing), in exchange for benefits such as relationship improvement or customized services [5]. Given its prevalence, importance, and possible negative consequences for users, PID has garnered increasing scholarly attention in information systems (IS) research. Recent studies have examined themes such as PID drivers [6–9], the "privacy paradox" [10], the intergenerational transmission of parental privacy concerns to adolescent self-disclosure [11], and the neural underpinnings of PID [12].

These behavioral PID studies are valuable, as they focus on human aspects of information privacy and individual privacy decisions and behavior, an ongoing research genre in the IS discipline [13]. However, it is important to note that humans, as social creatures, tend to be nested within hierarchical social structures (i.e., romantic partners, family groups, and work teams). Accordingly, their individual behavior, including PID behavior, is influenced by the social structures they are nested with, for instance, through group norms [14]. Prior behavioral PID studies, as we show later, have taken mostly an individual, self-centric view of information privacy. While this is an essential first step, it is important to acknowledge that information privacy is a multilevel concept [15–17]. Accordingly, there is a need to understand PID more systematically as a multilevel phenomenon [14], primarily for two key reasons. First, not only individuals but also social groups (i.e., multiple individuals with certain levels of shared understanding, goals, and purposes) can claim information privacy, develop norms, and influence the PID of individuals nested within those groups [14]. Second, an individual's information privacy is sometimes interconnected to that of other individuals within a social group—for instance, parents sharing their children's information on social media, or family pictures that contain their own and their children's information [18].

There have been attempts to start conceptualizing the multilevel nature of information privacy [1,3,14]. Such studies suggest that PID transcends the individual level when (1) the way individuals manage their personal information is subtly guided by the dynamics of the groups within which individuals are nested or (2) multiple individuals participate in the regulation and management of private information and hence claim information co-ownership, a perception of the extent to which people own and control information [19]. These conditions have become very prevalent. For instance, on social media, individuals' PID behavior may be influenced by their privacy benefit–risk considerations, but also by the norms dictated by the online or offline social groups to which they belong [20]. Similarly, when multiple people co-own the information, the sharing decisions may be influenced by considerations beyond their own benefits and risks [21]. Thus, there is a need to use a multi-stakeholder and multilevel perspective to examine PID [22]. By disregarding this perspective, we fail to unpack the intricate interplay

* Corresponding author.
*E-mail addresses:* Mingxin1@student.unimelb.edu.au (M. Zhang), Reeva.Lederman@unimelb.edu.au (R. Lederman), Oturel@unimelb.edu.au (O. Turel).

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between individuals' information disclosure behavior and the broader social dynamics that guide such behavior. Disregarding the multilevel perspective also limits practical insights, as it creates tunnel vision, emphasizing the needs, preferences, and concerns of individuals, rather than social groups [23,24].

Given the limited focus and studies on the multilevel nature of PID [14,22,23], we seek here to synthesize the foundation for such research and to inspire future multilevel PID research. We posit that to begin such a journey, a systematic review is needed. A systematic review serves to integrate previous separate streams of findings and drive an up-to-date understanding of where we currently stand and where we should be headed [25]. To this end, our review seeks to identify gaps in existing work and serves to put forth an agenda for future research on multilevel PID. This allows us to substantiate that by taking an individual-level perspective, prior IS research has largely overlooked the complexity of multilevel social structures (e.g., individuals nested within groups), which can lead to imprecise statistical results (i.e., ignoring dependencies among individuals in groups) and overlook important higher-level processes and factors.

We acknowledge that there have been several review articles and theoretical papers about multilevel information privacy [1,3,14,22]. This article builds upon this prior research and extends it in the following ways. First, we consolidate and delineate the multilevel nature of PID and build on reference disciplines to chart a path forward for the field. Second, we provide a comprehensive review of prior PID research in the IS discipline, aiming to identify specific opportunities for future multilevel PID research. Third, the review enables us to put forth novel propositions and demonstrate the need for multilevel analytical methods. Overall, our review is motivated by the ongoing call for multilevel PID research in the IS discipline [1,3,14,22]. However, little is known about whether and how prior IS studies have responded to this call, and if not properly responded to, how we can advance multilevel PID studies. As such, we aim to address the following research questions (RQ):

- RQ1: How can PID be viewed as multilevel in nature?
- RQ2: How have previous studies investigated it as a multilevel phenomenon?
- RQ3: How can future research conceptualize and study multilevel PID?

To address these questions, in Section 2, we first present key concepts and their definitions. This serves as a basis for the review process. Next, in Section 3, we present the review methodology, including literature search, inclusion and exclusion criteria, and analysis. Then we describe the results of the literature review (186 relevant papers) and synthesize them. Through this approach, we unpack why PID should be viewed as multilevel and how previous studies have investigated it as a multilevel phenomenon (Section 4). Based on this, we propose a multilevel theoretical framework and point to multilevel analytical tools that are needed for testing this perspective (Section 5). Last, in Section 6, we present our discussions, limitations, and concluding remarks.

## 2. Key concepts and definitions

In this section, we present the key concepts in our review article. This serves two main purposes. First, we clarify the scope of our study by establishing the boundaries of what constitutes PID literature (Section 2.1) and PID typologies (Section 2.2). Second, we refer to the definitions of multilevel phenomena (Section 2.3) and multilevel research (Section 2.4), which help us to develop a guidance framework for examining PID literature from a multilevel perspective. By outlining these key concepts, we aim to provide a holistic pre-understanding that informs both our literature search and literature analysis.

### 2.1. Private information disclosure

Private information disclosure (PID) is the voluntary sharing of sensitive personal information with others (e.g., online peers, online platforms, or service providers) in a digital environment [14,19,26]. The definition of PID enables us to clarify the boundary conditions of the review and points to relevant terminology (e.g., "self-disclosure" and "personal information disclosure") that can be used in the literature search (Section 3.1 and Section 3.2).

### 2.2. Information ownership

Information ownership refers to the extent to which individuals claim rights to and believe they exert control over information [14,19]. Importantly, information ownership can be interdependent and shared among multiple individuals. Hence, individuals not only claim ownership of their own information but also harbor varying levels of ownership of multi-party information [19]. Examples are a paper co-written by multiple authors and a picture that contains the faces of multiple people. Using this definition, we theorize a typology of PID in Section 4.1, based on how information ownership is distributed and shared among information co-owners.

### 2.3. Multilevel phenomena

A social phenomenon is multilevel when it spans several levels of theorizing and analysis; these include individuals residing in higher-level social units [27]. Hence, a multilevel phenomenon can be determined by (1) the existence of a hierarchical system of different social units and (2) the existence of dependence among concepts from these units [28,29].

Additionally, for multilevel IS phenomena, it is imperative to consider how technology is involved in the hierarchical structure of human social units [30]. This is because technologies represent an identifiable social component that exists on its own [31]. Thus, individuals are often nested within a combination of mixed-compositional models of human units and IT units [30]. For instance, an employee could be in a work team (humans) on Slack (technology). Hence, for multilevel research in the IS discipline, the focus is often on the mixed-compositional model (the left model in Fig. 1). This definition enables us to delineate the multilevel nature of PID (Sections 4.2 and 4.3), as well as to develop a theoretical framework to inspire future multilevel PID research (Section 5).

### 2.4. Multilevel research

Multilevel research examines the relationships between variables on at least two different levels of conceptualization and analysis, and accordingly requires data collection from different levels of units, the emergence of higher-level constructs, and cross-level analysis [27]. Hence, multilevel research can be identified through examining whether concepts encompass different levels of theorizing and whether multilevel analytical procedures are implemented, including multilevel data sampling, data aggregation, and cross-level analysis [29]. We employ this notion in Section 4.4, which enables us to identify whether and how multilevel PID research has been executed.

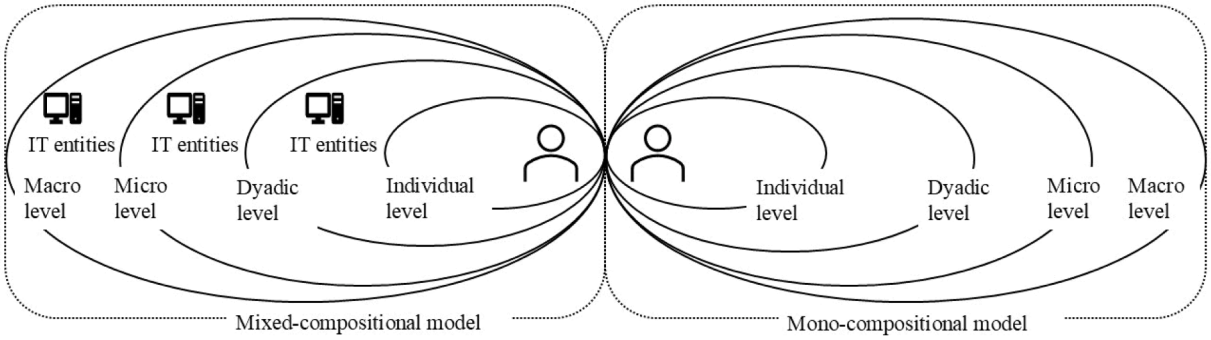



**Fig. 1.** Mixed-compositional and mono-compositional social structures.

## 3. Methodology

### 3.1. Keywords and selection of journals

To develop a search query, we first examined several relevant systematic review papers regarding information disclosure [32,33] and information privacy [1,3]. Using the keywords and ideas in these papers, we formulated the following search query:

- Abstract= ("information" OR "data" OR "privacy") AND Abstract= ("personal" OR "private" OR "self" OR "others" OR "co-owned" OR "group" OR "collective" OR "interdependent" OR "networked" OR "multiparty") AND Abstract= ("disclosure" OR "sharing" OR "providing" OR "disclose" OR "share" OR "provide" OR "give") AND SO= (Journal Title).

Because we aimed to contribute to the ongoing discourse on PID in leading journals, we restricted our search to the IS Senior Scholars' List of Premier Eleven Journals and to two widely recognized journal quality ranking lists: the ABDC journal ranking list (2022 version) and the ABS journal ranking list (2021 version). We targeted journals indexed under the information management category in the ABS journal list (tier 2, 3, 4, or 4*) and ensured that these journals were also categorized as either A-tier or A*-tier in the ABDC journal list. This (as opposed to accounting for non-listed journals or conferences) allows examining a manageable scope of ideas, and one that reasonably represent the state of the field [25]. In total, we had an initial pool of 34 journals. Using this journal pool, we applied the search query in Web of Science. We imposed no publication year limits.

We did not include Association for Information Systems (AIS) conference papers, for three reasons. First, there may be duplicate papers, as researchers may present their work at conferences first and then publish it in a journal outlet. Second, some conference papers represent in-progress ideas. Third, the AIS e-library search engine requires a different query format. To maintain a consistent approach in our search strategy, we chose to focus only on Web of Science. We recognize the possibility that our search strategy may have overlooked some relevant articles. However, our objective was not to cover all publications about PID. Instead, we sought to identify primary sources that could best reflect the current discourse within the IS community. Thus, we believe our search scope is adequate and serves as a good starting point for understanding the current state of PID research.

### 3.2. Search protocol: inclusion and exclusion criteria

Paper retrieval was conducted in two rounds. The first round was database search, where we used the search query to retrieve relevant papers in Web of Science. This process yielded 5070 papers, which were all reviewed (paper title, publication title, publication type, and abstract) at the screening stage. A paper was deemed suitable for further full-article assessments if it satisfied all the following criteria:

- The key phenomenon under examination was PID.
- The paper was written in English.
- Full-text access was provided.

This screening led to the removal of 3872 papers. Thus, 1198 full-article papers were further assessed. In the process of full article assessment, we read the full text of each paper and excluded papers if they met any one of the following criteria:

- They were technical research, corporate studies, review articles, opinion pieces, editorial papers, or meta-analyses.
- They were not quantitative studies.
- They were empirical studies with secondary data (i.e., no direct involvement of human participants).
- PID was not a behavioral outcome (i.e., a dependent variable).

Because we aimed to review empirical studies that explained the mechanism of PID, we removed technical research, theoretical papers, meta-analyses, and papers that did not deem PID an outcome variable. We acknowledge the importance of qualitative PID research. However, we removed qualitative research for two reasons. First, to maintain consistency in coding and analysis, we focused on quantitative research only. Second, multilevel research is typically quantitative, as it involves various quantitative procedures [30]. Importantly, for multilevel research, the focus is to understand how nested social structures influence individuals' behavior. Hence, behavioral research is inherently central to multilevel studies [27]. We thus excluded non-behavioral studies because they did not involve the collection of primary data from human participants. For instance, some studies explored PID using secondary data by scraping user posts and profile information to analyze how personal backgrounds influenced online self-disclosure [34].

After this first round, 168 papers remained. The second round aimed to analyze the references of the included papers. This process allowed us to identify additional papers, using the same inclusion and exclusion

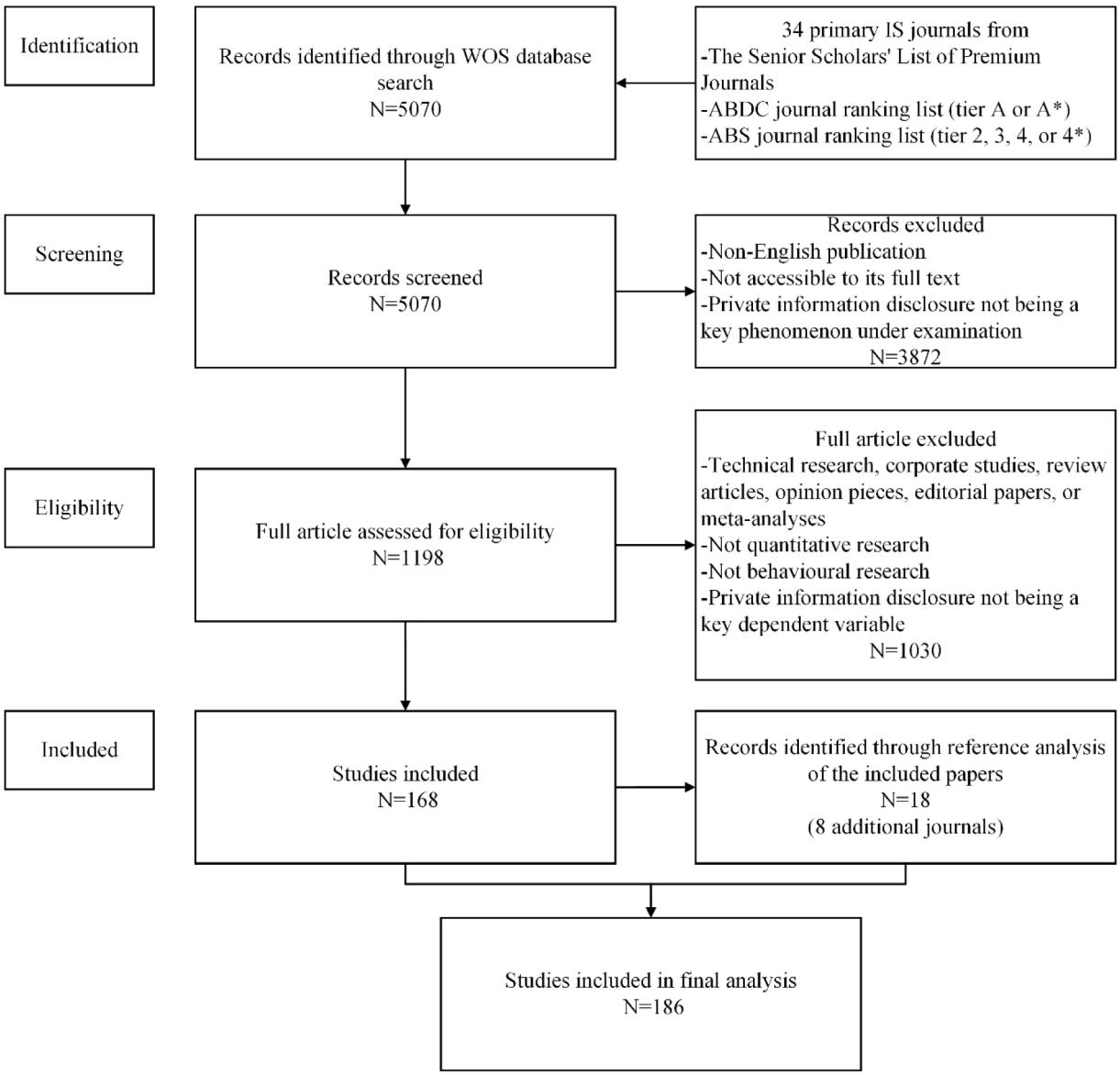



**Fig. 2.** Literature search.

criteria of the first round. The second round yielded another 18 papers indexed in eight additional journals, making the final sample 186 papers (Fig. 2). Seven IS journals at the database search stage did not yield a paper that met our exclusion and inclusion criteria.

### 3.3. Coding and analysis

We adopted a thematic approach to synthesize multiple studies to enable knowledge development from content analysis [32,35]. For each paper, we focused specifically on reviewing the introduction, hypothesis development, research model, methodology, and discussion sections. We developed a coding scheme and used a spreadsheet to manage the findings from the coding.

First, we coded their terminologies, definitions, information ownership, and measurement techniques. This afforded an understanding of how previous studies defined, conceptualized, understood, and measured PID. Second, we examined the theories or theoretical frameworks in each paper. By doing so, we were able to uncover the prior theorizing regarding PID. We also counted the frequency of each theory. Third, we extracted PID antecedents by looking at the hypotheses and the research model in each paper. For simplicity, independent variables, mediators, and moderators were deemed antecedents [32]. In addition, we coded the study context of each paper and teased out the specific IT artifacts mentioned in it. We also scrutinized the sample size, data collection methods, statistical methods, and level of analysis in each paper.

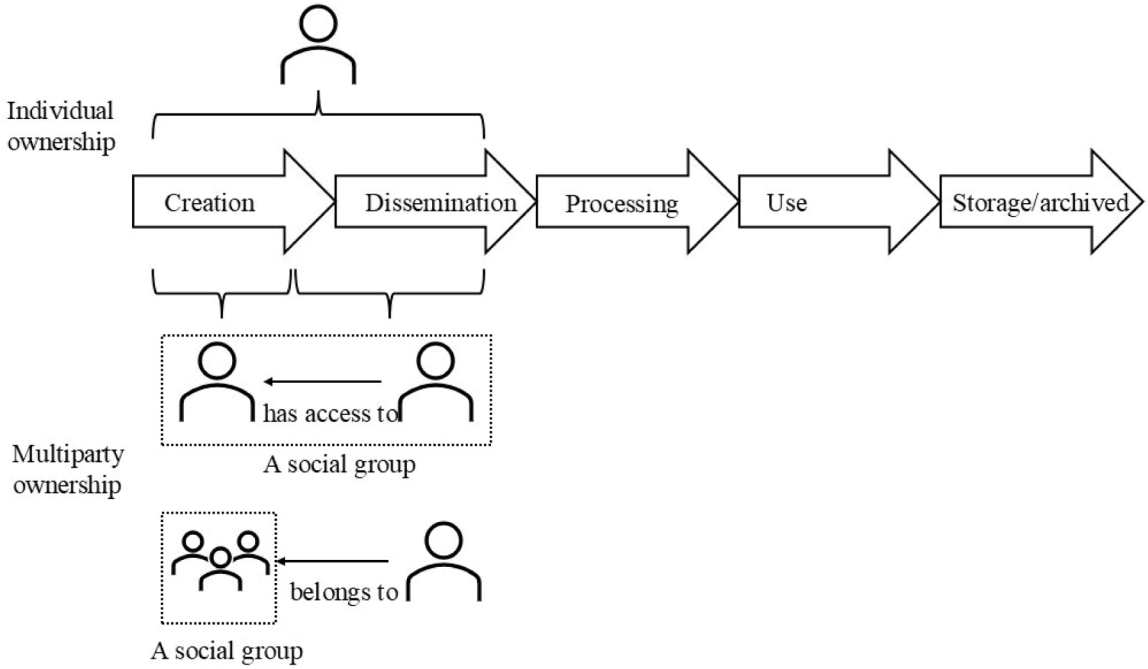



**Fig. 3.** A typology of private information disclosure.

## 4. Current state: private information disclosure across various digital contexts

In this section, we aim to address RQ1 and RQ2. For RQ1, we synthesized prior research on PID by unraveling its multi-stakeholder nature (Section 4.1), the theories that underpinned it (Section 4.2), and its antecedents (Section 4.3). For RQ2, we examined the methodology of prior studies (Section 4.4). We investigated if previous studies included procedures such as multilevel data sampling, data aggregation, and cross-level analysis [27].

### 4.1. The terminology: a typology of private information disclosure

Because private information sometimes encompasses personal details concerning multiple individuals, the act of disseminating such information extends beyond one individual and points to multiple stakeholders [3,22,23]. Based on the definition of information ownership (Section 2.2), we illustrate a PID typology, including self-disclosure, the disclosure of other people's information, and the disclosure of collectively owned information (Fig. 3).

#### 4.1.1. Individual ownership: self-disclosure

Self-disclosure refers to individuals' sharing of information about themselves [2,11]. Individuals create and disseminate their personal information, which they claim rights to and exert control over [19]. No other co-owners are involved at these two stages. Among the collected 186 papers, we found that most studies focused on individual ownership, exploring how and why people disclose private information about themselves and using terminologies such as "self-disclosure" or "personal information disclosure."

#### 4.1.2. Multiparty ownership: the disclosure of other people's information

Because information privacy is interdependent, the creation of private information involves one person, while the dissemination of it sometimes may capture different others [14]. An example is "sharenting, " describing how parents share their children's information on social media [18]. In the collected 186 papers, we found that multiparty ownership was less studied. Only six papers explored the disclosure of other people's information using terminologies such as "disclosure of

**Table 1**
The studied information ownership and disclosure behavior.

| Information ownership | Typology | Terminology | Number of publications | Sample article |
|---|---|---|---|---|
| Individual ownership | Self-disclosure | Self-disclosure | 57 | [36] |
| | | Intention/ willingness to disclose/provide/ share/give personal information/data | 53 | [37] |
| | | Information disclosure | 23 | [7] |
| | | Private information disclosure | 4 | [8] |
| | | Personal information disclosure | 3 | [38] |
| | | Other terminologies | 38 | [39] |
| Multiparty ownership | The disclosure of others' information | Disclosure of information about others | 1 | [40] |
| | | Exposure of others' information | 1 | [41] |
| | | Disclosure about others | 1 | [42] |
| | | Willingness to share data about others | 1 | [43] |
| | | Intention to share others' information | 1 | [44] |
| | | Self's sharing of information ownership | 1 | [45] |
| | The disclosure of collectively owned information | Picture sharing preference | 1 | [46] |
| | | Co-owned information disclosure | 1 | [47] |

information about others" or "disclosure about others."

#### 4.1.3. Multiparty ownership: the disclosure of collectively owned information

Multiple stakeholders within a group can co-generate and co-regulate private information [14,19]. An example is an individual who uploads a family group photo on Facebook. Such multiparty ownership is unique because (1) it captures multiple co-owners at the information creation and dissemination stage and (2) the co-owner who disseminates the information also participates in the information creation stage. Among the collected 186 papers, we found two papers examining the disclosure of collectively owned information. Kim and Gweon [46] studied picture owners' sharing preferences when (1) their relationship closeness with other people captured in the picture was high versus low and (2) the number of people captured in the picture was large versus small. Feng et al. [47] adopted cross-sectional surveys to examine how self-calculus, others' calculus, and collective calculus could jointly predict co-owned information disclosure on social media (Table 1).

#### 4.1.4. A brief summary

By examining the PID typology, we confirmed the multi-stakeholder nature of PID. Indeed, PID often transcends the individual level because shared information can capture personal details concerning multiple people. We specifically identified three phenomena: (1) self-disclosure, (2) the disclosure of others' information, and (3) the disclosure of collectively owned information. Our analysis showed that prior IS studies have mostly examined self-disclosure, while relatively overlooking the other two forms of disclosure. This represents an important

**Table 2**
Foundational perspectives on information privacy and the underlying social units.

| Foundational perspectives | Social units | Explanations | Key perspectives | Primary theories | Frequency |
|---|---|---|---|---|---|
| Privacy as a commodity | Individual level - Single person | Disclosure as personal cost-benefit trade-offs | Assessment of privacy benefits and privacy risks | Privacy calculus theory | 71 |
| | | | Evaluation of costs and benefits while engaging in social relationships | Social exchange theory | 9 |
| | | | Evaluation of gains and potential losses while making decisions | Prospect theory | 2 |
| | | | The pursuit of rewards and the avoidance of punishment | Incentive theory | 1 |
| | | Extension: Inhibitors and drivers of the cost-benefits trade-offs in the decision-making process | Rational deliberation and its impacts on people's attitudes and behavioral intentions | Theory of reasoned action | 10 |
| | | | Rational deliberation and its impacts on individuals' attitudes and perceived behavioral control | Theory of planned behavior | 6 |
| | | | Threat appraisal and coping appraisal | Protection motivation theory | 7 |
| | | | The central route and the peripheral route | Elaboration likelihood model | 3 |
| | | | The intuitive system and the analytical system | Dual process theory | 2 |
| | | | Two motivational systems: the prevention system and the promotion system | Regulatory focus theory | 2 |
| | | | Positive affect and negative affect | Affect heuristic theory | 1 |
| | | | Psychological distance | Construal level theory | 1 |
| Privacy as control | Interpersonal level - Dyad | Disclosure as a form of selective control of others' access to the self in interpersonal communication | Boundary regulation in managing interpersonal interaction | Communication privacy management theory | 13 |
| | | | The value of social relationships in interpersonal communication and interaction | Social capital theory | 7 |
| | | | The perceived overlap of one's self-concept with that of others during interpersonal communication | Self-construal theory | 3 |
| | | | The exchange of personal information that develops and deepens interpersonal relationships over time | Social penetration theory | 2 |
| | | Extension: Technology-mediated interpersonal communication | Technology use in promoting relationship fulfillment | Use and gratification theory | 3 |
| | | | How a sense of being with others is mediated by digital interfaces | Social presence theory | 2 |
| | | | Affordance and user interface design in enabling online virtual interaction | Technology affordance theory | 1 |
| | | | Behavioral intervention by technology design in online interpersonal communication | Nudge theory | 1 |
| Privacy as a subjective state | Micro group level - Social group | Disclosure as a form of compliance with privacy norms/rules associated with a social role or group identity | Social role and its corresponding responsibilities and social expectations | Social role theory | 5 |
| | | | Group membership in guiding attitudes and behaviors | Social identity theory | 2 |
| | | | Privacy norms in a specific context underpinned by contextual factors | Contextual integrity theory | 4 |
| | Macro group level - Culture and society | Disclosure as a form of compliance with privacy norms/rules ingrained in culture and society | Cultural differences and their impacts on privacy perceptions | Hofstede's cultural dimension framework | 4 |

research opportunity and informs propositions for future research, on which we systematically elaborate in Section 5.1.

### 4.2. Theory: A hierarchical system of social units behind private information disclosure

Nested within a hierarchical social structure, higher-level social units usually confer implicit or explicit effects on individuals, rendering their perceptions, beliefs, and behavior subtly guided by a broader social dynamic [48]. Using the definition of multilevel phenomena (Section 2.3), we argue that individual PID can be deemed multilevel because it occurs within a hierarchical social structure consisting of different levels of human social units. In that section, we examined the theories in each collected paper to elicit the human social units behind individual PID. Theories are proposed structured explanations of why and how a particular social phenomenon occurs [49]. By looking at theories pertinent to a social phenomenon, its driving forces and components can be elicited.

As shown in Table 2, we found that individual PID is an interpersonal communication process that is nested within social groups and bound by societal and cultural influences. Hence, PID is not only an individual-level phenomenon; instead, it is influenced by and associated with higher-level social units. Additionally, empirical individual PID studies are grounded in three foundational perspectives on "what is privacy" [3]. We resorted to these perspectives and theoretically mapped them onto different levels of human social units. Through this approach, we argue that, when studying individual PID behavior, the meaning of "privacy" is contingent upon the social unit it is associated with. Below we elaborate.

#### 4.2.1. Individual level: a cost-benefit trade-off and its inhibitors and drivers

Information privacy can be deemed a "commodity" [3,50]. According to this perspective, individuals are primarily driven by the net value of their mental cost-benefit trade-offs [50]. When individual PID behavior is studied, this "privacy as a commodity" perspective corresponds to the individual level, as it focuses on personal cost-benefit analysis and its effects on individuals' PID decisions. Theories supporting this body of work include privacy calculus theory, social exchange theory, prospect theory, and incentive theory. Moreover, human beings have finite cognitive resources and cannot invariably make informed decisions [50]. Prior research has extended the cost-benefits trade-off approach by incorporating impulsive, non-rational, or heuristic factors, showing how the mental calculus may be swayed or biased [10,51]. Theories supporting this body of work include affect heuristic theory and dual process theory. Overall, viewing PID as an individual-level phenomenon, prior research has mostly deemed privacy a commodity that individuals can trade off, influenced by such individual factors as personal biases [52] and mood states [10].

#### 4.2.2. Interpersonal level: interpersonal communication and its extension in a technology-mediated environment

Information privacy can be deemed a form of "control" over information flow in interpersonal communication [3]. Through this control, people navigate between public and private spheres and decide when and how to engage in social interactions [19]. As such, PID represents a dyadic and interpersonal communication process, which cannot occur in the absence of information recipients or audience [19]. Thus, the "privacy as control" perspective corresponds to the interpersonal level or the dyadic level, as it focuses on how individuals engage in interpersonal communication by controlling the width and depth of information exchanges between others and themselves [53]. Theories supporting this body of work include communication privacy management theory, social capital theory, and social penetration theory. In addition, in a digital environment, interpersonal communication is mediated and facilitated by IT artifacts, where various forms of control (e.g., privacy settings and privacy nudges) can be leveraged to manage information exchange and communication between others and oneself [5]. Theories supporting this body of work include social presence theory, affordance theory, and use and gratification theory. Overall, viewing PID as an interpersonal-level phenomenon, prior research has studied how individuals exercise control over information flow in interpersonal communication (e.g., sender–recipient relationships), facilitated by IT artifact affordances.

#### 4.2.3. Micro group level: privacy norms associated with a social identity in each context

Information privacy can also be deemed a subjective "state" [3,51], often achieved when individuals behave properly in alignment with the normative expectations rooted in their salient social identities [14,26]. To explain, a social context fosters a salient social identity, referring to an individual's self-concept derived from their membership in a social group [14]. This salient social identity is important, as it can dictate privacy norms, regulating what information should be shared with whom and under what circumstances [26]. Individuals act on those normative beliefs, thereby navigating their information communication and exchange in the focal context [14]. When information flows in alignment with privacy norms, it enables individuals to yield a subjective state of "privacy," namely perceived appropriateness of information flow in a given context [26]. For example, disclosing personal drug use to a physician (in the role of a patient) is seen as more appropriate than sharing it with a colleague (in the role of a co-worker). Indeed, humans navigate a wide range of social contexts, and in each context, they play a different role and have distinct expectations regarding how to manage information flow appropriately in that context [54]. Consequently, their "state" of privacy is often fluid, depending on the most salient identity and the privacy norms available to them that shape their privacy experience. Hence, the "privacy as a state" perspective corresponds to the micro group level, as it focuses on how individuals retrieve and comply with privacy norms associated with their salient group identities in a given context. Theories supporting this body of work include social role theory, contextual integrity framework, and social identity theory.

#### 4.2.4. Macro group level: a cultural and societal phenomenon

Cultural identity, as a macro-level social identity, can shape people's perceptions of information privacy, meaning that the notion of "information privacy" may be interpreted differently across cultures [55]. Hence, the "privacy as a state" perspective can also apply to the macro group level, as individuals hailing from distinct cultures may have culture-specific norms regarding the regulation and management of private information. Accordingly, individual PID is intertwined with cultural norms and cannot stand alone in the absence of the cultural and societal influences that drive it [56]. Thus, from a cultural and societal perspective, a subjective state of "privacy" can be achieved if culture-specific privacy norms are respected. In this regard, despite not being typically deemed a theory, Hofstede's cultural dimension framework has been used widely to explore how cultural and societal factors (e.g., uncertainty avoidance, individualism and collectivism) could shape people's perceptions of information privacy, thereby driving their PID decisions [39,41,57].

#### 4.2.5. A brief summary

Analyzing and synthesizing the theories employed in prior studies,

we point to three foundational perspectives that can be mapped onto different levels of human social units. Through this approach, we unpack the multilevel nature of PID and show that individual PID behavior could potentially occur in a nested social structure.

A key insight that emerged from our analysis is that these foundational perspectives for information privacy are not mutually exclusive, and in fact, they are sometimes intertwined and interrelated. Notably, we observed an important paradigm in the literature, where the "commodity" perspective is often extended by incorporating elements of the "control" and/or "state" perspectives. Indeed, while the "commodity" perspective focuses on individual cost-benefit evaluations (the privacy calculus), such assessments are frequently influenced by individuals' perceived control over their data flow and the broader social contexts in which individual PID decisions are made. For instance, when PID occurs, a sense of control over personal data flow can reduce privacy risks [58]. In addition, individuals' cost–benefit assessments may vary depending on the circumstances in which PID decisions are being made [26], including the roles individuals play in that moment [54]. This implies that the "state" perspective can also influence the "commodity" perspective, as individuals' mental cost-benefit trade-offs can be dynamic and contingent on contextual and role-based factors.

Despite these insights, the "commodity" perspective still needs to be extended properly, particularly in the following two ways. First, PID typically occurs within sender–recipient relationships, where individuals act as senders who exercise selective control over their own data [17,26]. However, in a networked digital environment, individuals can control not only access to their own data but also access to others' data [14]. When individual PID occurs in sender–subject relationships, a critical question emerges of how individuals perform mental calculi when controlling others' data. This informs propositions for future research, which we discuss systematically in Section 5.2.1.

Second, social groups provide contexts in which individuals are

**Table 3**
The studied antecedents for private information disclosure.

| Entities | Level of units | Antecedent group | Specific studied antecedents | Sample article |
|---|---|---|---|---|
| Human entities | Individual level | Personality traits | • Extraversion<br>• Agreeableness<br>• Conscientiousness<br>• Narcissism | [40] |
| | | Previous experiences | • Privacy invasion experiences<br>• Privacy disclosure experiences | [61] |
| | | IT use | • Social media literacy<br>• Technology usage frequency<br>• SNS use | [62] |
| | | Perceived benefits | • Social benefits<br>• Hedonic benefits<br>• Utilitarian benefits | [63] |
| | | Perceived risks | • Privacy risks<br>• Perceived risk severity<br>• Perceived risk susceptibility | [41] |
| | | Privacy concerns | • Information privacy concerns<br>• Interaction privacy concerns<br>• Psychological privacy concerns<br>• Parental privacy concerns | [64] |
| | | Affect/emotions | • Affect<br>• Joy<br>• Fear<br>• Mood state | [65] |
| | | Time perspective | • Distant-near future intention | [66] |
| | | Heuristics | • Risk seeking/aversion<br>• Impulsivity<br>• Habit<br>• Laziness | [67] |
| | Interpersonal level | Relational consideration | • Reciprocity<br>• Relational trust<br>• Interpersonal closeness | [68] |
| | | Recipient/audience impact | • Peer pressure<br>• Parental mediation | [69] |
| | Micro group level | Group characteristics | • Group centrality<br>• Group size | [70] |
| | | Group membership | • Role conflict<br>• Group identification<br>• Group cohesion | [54] |
| | | Group norms | • Norms of reciprocity<br>• Norms of shared information ownership | [45] |
| | Macro group level | Individualism/collectivism | • Individualistic cultural orientation<br>• Collectivistic cultural orientation | [41] |
| | | Uncertainty avoidance | • Uncertainty avoidance | [71] |
| IT entities | | Information attributes | • Information sensitivity<br>• Information type<br>• Data security | [72] |
| | | Information system design | • Privacy seals<br>• Social cues<br>• Anthropomorphic design | [73] |
| | | Information services and functions | • Privacy policy<br>• Privacy settings<br>• Privacy control | [74] |

situated [59], thereby subtly shaping individual PID decisions [26]. However, as outlined earlier, information privacy is a multilevel concept [3], meaning that groups can claim privacy and treat it as a shared resource (i.e., a commodity) that can be traded off [60]. In this sense, the "commodity" perspective also needs to be extended to the group level, by considering the processes through which groups perform privacy calculi and make group PID decisions. This aspect, however, remains largely under-explored, which we discuss systematically as propositions for future research in Section 5.2.2.

### 4.3. Antecedents: the dependence among concepts from different social entities and social units

Our analysis in Section 4.2 has revealed that individual PID behavior could occur in a nested structure of social units. While this social structure mostly points to human entities (e.g., individuals, dyads, and groups), it is also important to note that, according to Belanger et al. [30], individuals are often nested within a combination of mixed-compositional models. Using the definition of multilevel IS phenomena (Section 2.3), a multilevel social phenomenon encompasses dependence among concepts from different levels of social units, including those of (1) human entities and (2) IT entities. Hence, IT entities should be viewed as an independent social unit that exists on its own, usually embedded in multilevel human social structures (e.g., couples nested within a dating app) and not just at one single level. In this section, we identified the antecedents of PID and organized them by type of social entity and unit (Table 3), thereby separately reviewing and examining the effects of human units and IT units on PID. In doing so, we not only further confirmed the multilevel nature of PID but also revealed that PID could indeed occur in a mixed-compositional model.

#### 4.3.1. Individual-level antecedents

Most current research has focused on individual-related antecedents, employing the calculus lens and extending it by considering individual factors such as personal background (e.g., demographics and personality traits; [75]), technology use [62], and more notably, individuals' impulsive decision-making tendencies [54]. Indeed, human beings have finite cognitive resources, rendering their mental calculus and subsequent sharing decisions bound by non-rational factors and mental shortcuts [50]. Prior studies have examined this idea, showing that individual PID could potentially be biased or dominated by habits [76], temporal factors [12], mood states [10,65], or heuristics [66,77].

#### 4.3.2. Interpersonal-level antecedents

Some studies have also adopted an interpersonal-level perspective, examining how the interpersonal communication between information senders and information recipients can drive PID. For example, past research showed that adolescents adjusted their disclosure strategies based on friend types and relational ties, through which they decided what information should be shared with whom under what circumstances [62]. Within the family context, parental privacy concerns can pass down to adolescents, and through this mechanism, adolescents may be less inclined to self-disclose online [11]. In addition, the characteristics of interpersonal relationships (e.g., relationship intimacy, closeness, and relational trust) between information senders and information recipients are also vital for PID [5,26]. For example, individuals are more likely to disclose sensitive information to intimate others because they seek to consolidate interpersonal relationships and obtain social support from close social networks [78].

#### 4.3.3. Micro-group-level antecedents

Social groups within which individuals are nested can have implicit or explicit effects on individual PID, usually manifested by group identification, group norms, or group characteristics. First, group identification is defined as the part of individuals' self-concept that derives from their group membership perceptions [79]. Past research has shown that being a valuable group member increases individuals' privacy concerns toward other intragroup members, and consequently reduces their intention to disclose others' private information [40]. Second, group norms refer to a set of behavioral rules concerning the shared understanding of acceptable behavior that individuals learn from other group members [14]. Individuals navigate the most salient group identity available to them in a given context and make PID decisions based on the corresponding group norms [14]. Last, group global characteristics such as group size could also affect individual PID [1,68]. For example, information sharing in larger groups is deemed riskier, and individuals nested within such groups are more reluctant to disclose private information [70,76].

#### 4.3.4. Macro-group-level antecedents

As a macro-level factor, cultural factors may shape individuals' perceptions of information privacy, thereby affecting their PID decisions [39]. For example, individuals in individualistic cultures prioritize personal goals, achievement, and enjoyment, leading them to be more influenced by hedonic motives when making PID decisions [41,57]. Uncertainty avoidance refers to the extent to which people feel threatened by ambiguous and unpredictable events [80]. Uncertainty avoidance is closely related to risk perceptions. Past research has shown that people in high-uncertainty-avoidance cultures are more wary of privacy risks and less willing to share private information [71].

#### 4.3.5. IT entity-related antecedents

To unravel IT-related effects, we leveraged ideas from the information systems success model [81,82], and categorized IT entity-related antecedents into the following three dimensions. First, information attributes refer to the formats, quality, and sensitivity of personal data shared via information systems [81]. Information attributes are an important factor to consider, as they affect how individuals perceive privacy risks and manifests privacy concerns [72]. Second, IT design characteristics could nudge information-sharing behavior, as individuals' privacy reflections may be nullified or inhibited by visual cues or privacy-related nudges [73,83]. For example, security seals [84] and social cues [8] have been found to afford mental shortcuts, leading individuals to disregard privacy concerns and disclose personal information in a more impulsive manner. Last, IT artifact services and functions could also affect individual PID. Prior research has examined how privacy settings [85], privacy controls [74,86], and privacy policies or assurances [4] could bolster self-efficacy, perceived control, and trust, as well as reduce privacy concerns. Digital platforms leveraged these functionalities to increase users' willingness to provide personal data [87].

#### 4.3.6. A brief summary

Resorting to the definition of mixed-compositional models, Section 4.3 complements Section 4.2 by categorizing human and IT entities separately to differentiate their effects on PID. Through this approach, we unpacked the complexity of PID by specifically illustrating how it could be affected by multilevel social units of both human entities and IT entities, thereby further unpacking the multilevel nature of PID. We showed that IT entities represent an independent social unit in parallel to human social units. In particular, IT-afforded control can subtly shape

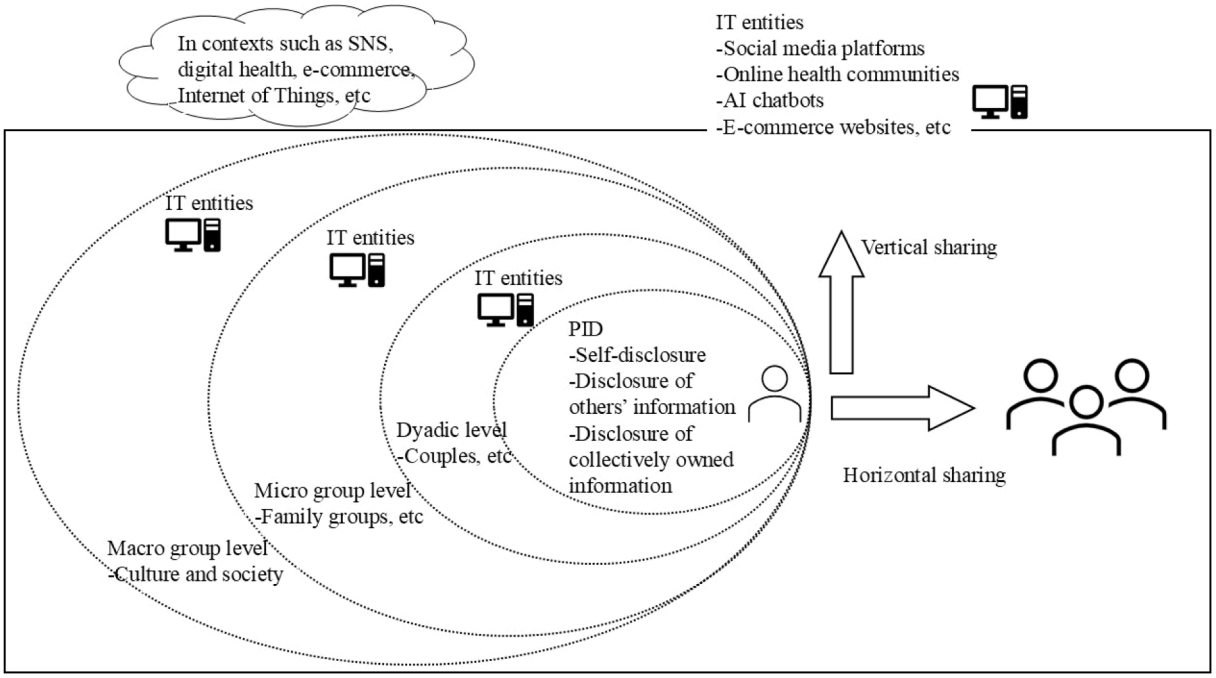



**Fig. 4.** The multilevel nature of private information disclosure across contexts.

individuals' mental cost-benefit trade-offs, thus driving their PID decisions. For example, social media platforms afford various forms of privacy settings, which individuals can leverage to reduce their privacy risks, enhance trust, and ultimately manage their data flow according to their preferences [74,86].

Nonetheless, IT entities are not merely embedded in one level of human social unit, but span multiple human social units, a key characteristic of mixed-compositional models [30]. Existing research has mostly focused on how IT-afforded control can bolster perceived privacy in interpersonal communication and shape individuals' mental calculi when they make PID decisions. This overlooks how IT-afforded control can also influence the way higher-level human social units (e.g., groups) perform mental calculi and make group PID decisions [60,88]. Notably, emerging IT artifacts such as big data analytics extend into the realm of group privacy, where individuals can be profiled into different groups based on their demographics, consumption history, or personal interests [88]. This trend may potentially increase group privacy threats, signifying the need for effective group privacy management in an ever-evolving IT-nested social structure. This gap in the literature informs propositions for future research, on which we elaborate in Section 5.2.3.

Taking these together, we addressed RQ1 in Sections 4.2 and 4.3. Our literature synthesis in these sections confirmed the multilevel nature of PID. This is because (1) "private information" sometimes encompasses personal details concerning multiple stakeholders, and hence the sharing of such information extends beyond the individual level, and (2) individuals are nested within hierarchical systems of human social units (e.g., families, work teams, social media friends) and IT units (e.g., users of an app), and their PID behavior could potentially be affected by such a hierarchical social structure. As shown in Fig. 4, PID occurs in a wide range of contexts, such as SNS, e-commerce, digital health, and the Internet of Things, via mobile apps, wearable technologies, social networking sites, and so forth (see Appendix D). Within these contexts, individuals often reside in a mixed-compositional social structure consisting of both human units and IT units, making their PID behavior subject to such complex dynamics.

### 4.4. Methodology: multilevel data collection and analysis

In this section, we aim to address RQ2. According to the definition of multilevel research (Section 2.4), a multilevel phenomenon entails multilevel data analytical approaches to yield more nuanced insights [89]. Failing to do so may result in an oversimplified understanding of the complexities inherent in individuals' decision-making processes, leading us to overlook group-level dynamics and the effects of higher-level units. Hence, we examine if previous PID research has followed the typical multilevel analysis procedures: sampling multilevel data, aggregating individual data, and executing cross-level analysis [30].

#### 4.4.1. Multilevel data sampling: the unit of data collection

Empirical data can be collected at the individual level, the dyadic level, or the group level [27]. In multilevel research, it is crucial to collect data beyond the individual level, meaning that each data unit should involve input from at least two individuals. This indicates two types of multilevel data: dyadic data (i.e., two individuals as one data unit) and group data (i.e., three or more individuals as one data unit).

Dyadic data refer to paired observations obtained from two individuals who are interconnected, often within a social relationship (e.g., a couple, colleagues, or friends) [90]. Dyadic data are often collected and analyzed to unpack the interpersonal mechanisms in various types of dyads such as husband–wife, teacher–student, manager–employee, and parent–child [90]. Among the collected 186 papers, we found that most collected individual data, while only four studies collected dyadic data [9,11,70,91]. For example, in a study exploring how parents' privacy concerns influenced adolescents' online self-disclosure [11], researchers first reached out to high-school students, providing students with a paper-based questionnaire to measure their online self-disclosure

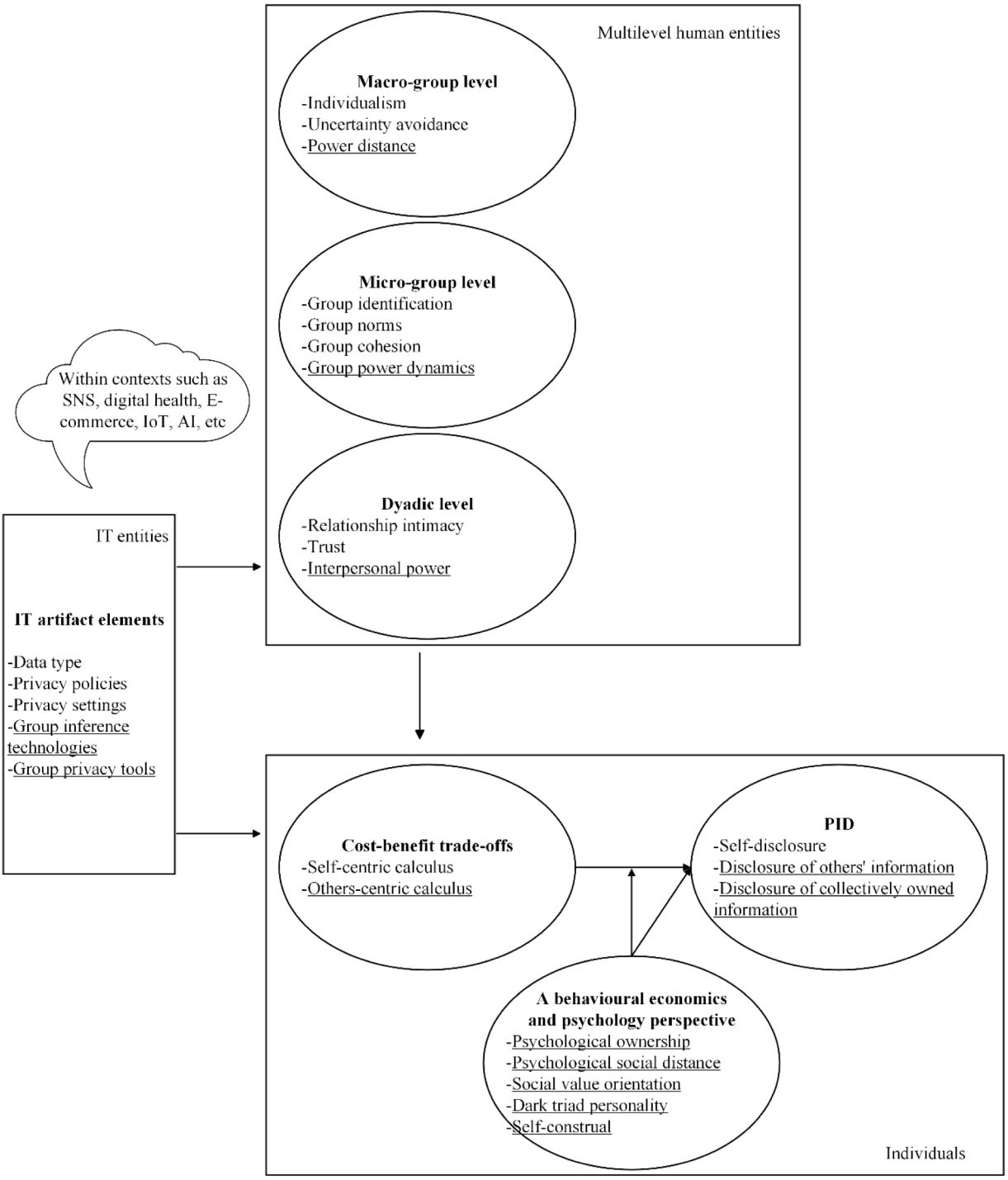



**Fig. 5.** Proposed multilevel research on private information disclosure across contexts *Note:* The underlined text denotes opportunities for future research, elaborated in the following sections. The listed bullet points are not exhaustive.

behavior. Then students were required to take separate questionnaire surveys to their parents, which measured parents' privacy concerns. Each student's response was paired with their parents' response for further analysis.

Group data refers to observations obtained from multiple individuals as one data unit [89]. For example, a family could be a unit of data collection, where researchers can combine responses from all family members to derive insights into family groups. Applying this idea to privacy management, data collected from family groups could potentially afford insights into family privacy management [19]. Among the 186 papers, we identified one paper that collected and analyzed group data [68]. In that study, 378 individuals nested within 42 online virtual communities were recruited, who generated group-level constructs and explored how group dynamics influenced individual members' online self-disclosure [68].

#### 4.4.2. Multilevel data analysis: collective constructs, data aggregation, and cross-level analysis

Multilevel data analysis typically follows three steps: the emergence of collective constructs, data aggregation, and cross-level analysis [27].

The first step is to identify or generate collective constructs. Collective constructs refer to the characteristics of social units at higher levels, in contrast to individual constructs at the individual level [27]. There are three types of collective constructs: *global constructs, shared constructs*, and *configurational constructs* [27]. *Global constructs* are objective group attributes, which are usually represented in the form of descriptive characteristics such as group size [89]. Global constructs do not derive from individual-level attributes and therefore do not require data aggregation or justification [27]. Among the collected 186 papers, we did not identify a study explicitly employing global constructs. *Shared constructs* can be established when individuals in a group have a shared understanding of a group-level attribute. For instance, the quality of a group leader gets its meaning as a team-level construct only when team members have reasonable agreement regarding the value of this concept. Among the collected 186 papers, we identified one paper involving shared constructs. This study conceptualized reciprocity, relational trust, and group identification as shared understandings among all members [68]. *Configurational constructs* are usually derived in a more complex and non-linear way, meaning that some individuals may contribute more to a group-level characteristic than others [27]. Among the collected 186 papers, we did not identify a study utilizing configurational constructs.

The second step is to justify the quality of data aggregation statistically. The justification for the emergence of collective constructs (i.e., the levels of agreement among group members on the focal construct) should be supported by assessing within-group agreement and between-group variability [89]. This can be demonstrated through measures such as inter-rater agreement and inter-rater reliability, respectively [89]. Inter-rater reliability and inter-rater agreement can be established by examining intraclass correlation coefficients (ICC), within-group agreement for a single item ($r_{WG}$), and average deviation indices. Out of the 186 papers, we found only one paper using $r_{WG}$ and ICC to justify data aggregation [68].

The third step is to perform cross-level analysis. Multilevel statistical techniques should be employed to investigate the relationships among constructs at different levels, including cross-level direct, cross-level mediation, and cross-level moderation effects [48]. There are two main techniques for cross-level analysis: hierarchical linear modeling (HLM) and multilevel structural equation modeling (MSEM). Only one paper in our sample pool utilized HLM as a research tool to explore the cross-level direct effects and cross-level moderation effects [68].

Here we use Lee et al. [68] as an example to demonstrate multilevel data analysis procedures. In their study, the authors aimed to investigate how group-level factors, such as group-level trust, identification, and norms of reciprocity, could influence individuals' self-disclosure in online communities. They first measured each individual's perceptions of trust, identification, and norms of reciprocity within their respective groups. Then individuals' responses (to each measurement item) within each group were averaged to generate group-level constructs for each corresponding group. The authors used $r_{WG}$ and ICC to assess the reliability of these group-level constructs, confirming that the indices were within an acceptable range. Finally, they applied hierarchical linear modeling (HLM) to examine cross-level direct and moderating effects.

#### 4.4.3. A brief summary

The multilevel nature of PID necessitates a broader and careful empirical focus on groups in addition to that on individuals [88,92]. Regardless, analyzing whether and how prior studies have implemented multilevel analytical procedures, we found that most empirical studies used individuals as the unit of data collection and data analysis. This indicates that existing studies were predominantly focused on the individual level rather than multilevel. Thus, we have had limited understanding of the multilevel PID mechanism, signifying a pressing need for future research in this domain. This informs important propositions for future research, which we systematically demonstrate in Section 5.3.

## 5. Future agenda: an integrated framework for multilevel research on private information disclosure

In this section, we aim to address RQ3. To inspire high-quality future multilevel research on PID, we recognize the importance of (1) a multilevel theoretical framework that guides it and (2) introducing multilevel analytical tools pertinent to testing multilevel theories. First, we develop a multilevel theoretical framework with directional influences depicted using arrow lines (see Fig. 5). In this framework, we distinguish clearly between established knowledge and areas requiring further investigation by using underlined text. The listed bullet points in Fig. 5 are not exhaustive; they are intended to illustrate examples of previously studied constructs and highlight under-studied yet potential constructs for future research. Second, to illustrate how Fig. 5 can guide future research, we present Table 4 to summarize the limitations of previous research and propose a future research agenda. This allows us to consider the opportunities presented by limitations in previous research. We also introduce the main procedures of conducting multilevel research, together with multilevel analytical tools and examples pertinent to multilevel PID research. We elaborate in the following sections.

### 5.1. The terminology: beyond self-disclosure

Past IS research has concentrated predominantly on self-disclosure, whereas multi-party information disclosure has been comparatively less studied. The disparity in scholarly attention signifies a pressing need to progress toward information disclosure pertaining to multiple stakeholders (e.g., a group photo or a screenshot of an online chat). This is because much of the disclosure on digital platforms such as social networking sites, online health communities, and e-commerce websites pertains to multi-party information [14,24].

An understanding of multi-party information disclosure can provide practical implications regarding multi-party privacy protection, inspiring practitioners to design functions and services pertaining to multiple stakeholders. Moreover, there are nuanced differences in the

**Table 4**
Previous limitations and future agenda.

| Section | Limitations of previous studies | Opportunities for future research | Propositions | Possible predictors |
|---|---|---|---|---|
| 5.1. The Terminology | Largely focusing on individual information ownership and self-disclosure | Future studies can explore multi-party information ownership, explaining the underlying mechanism of the disclosure of others' information and that of collectively owned information. | • P1: Why and how do people share private information about others?<br>• P2: Why and how do people share private information that is collectively owned with others?<br>• P3: What are the differences in the underlying mechanisms between sharing others' information and sharing collectively owned information? | • Psychological ownership<br>• Psychological social distance |
| 5.2. Theory | Largely focusing on a self-centric calculus approach | Future research can adopt an others-centric calculus approach, exploring whether and how individuals use perspective-taking to consider the benefits and risks to others when sharing multi-party information. | • P4: How do individuals use an others-centric calculus approach when they share multi-party information?<br>• P5: What are the factors that move privacy calculus from being self-centric to others-centric?<br>• P6: How do we compare the effects of the self-centric calculus and others-centric calculus? | • Social value orientation<br>• Dark triad personality<br>• Self-construal<br>• Interpersonal power |
| | Largely focusing on privacy calculus at the individual level | Future research can extend the individual-level privacy calculus by adding group-level extensions or understanding how higher levels of social units adopt privacy calculus. | • P7: What are the group-level factors that can influence individuals' mental calculus?<br>• P8: How do group-level factors influence individuals' mental calculus and consequently result in PID at the individual level?<br>• P9: How do contextual factors influence the effects of group-level factors on individual calculus?<br>• P10: How do different types of social groups apply the calculus lens at the group level?<br>• P11: How does group calculus influence PID at the group level? | • Group norms<br>• Group cohesion<br>• Group identification<br>• Group power dynamics<br>• Societal power distance |
| | Overlooking the role of IT entities in shaping how social groups utilize the privacy calculus approach | Future research can explore how IT characteristics influence not only individual calculus but also group calculus. | • P12: What are the emerging IT characteristics that can influence the way social groups use the calculus lens and make PID decisions?<br>• P13: How do emerging IT characteristics influence the way social groups use the calculus lens and make PID decisions? | • Group privacy tools<br>• Group inference technologies |
| 5.3. Methodology | Mostly collecting individual-level data and performing single-level analysis | Future research can aggregate individual data, generate collective constructs, and use multilevel analytical tools to execute cross-level analysis. | • P14: How do we use indices such as intraclass correlation coefficients to justify between-group variability and within-group agreement when conducting multilevel PID research?<br>• P15: How do we use multilevel data analytical tools to conduct multilevel PID research? | |

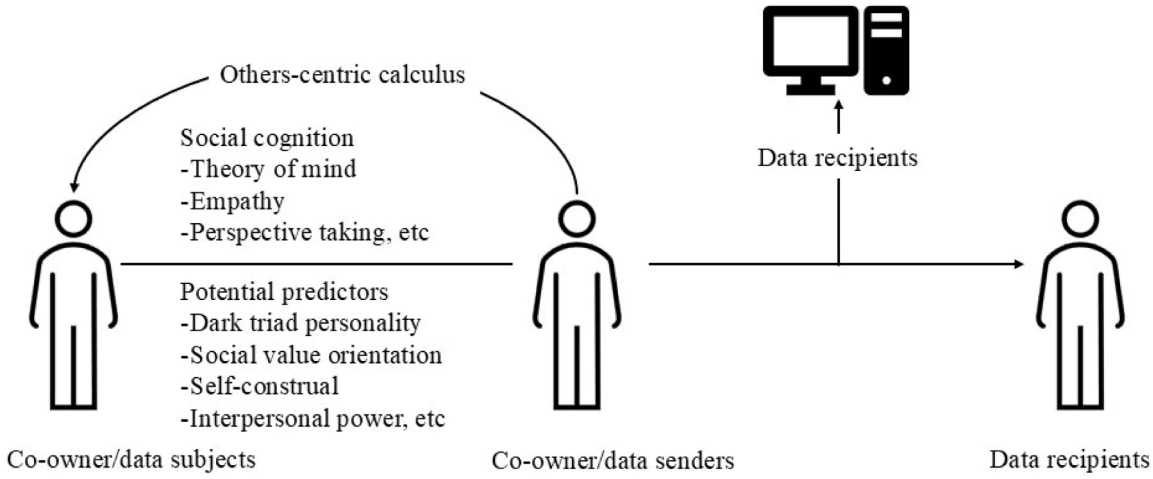



**Fig. 6.** The others-centric calculus and its potential predictors.

theoretical mechanisms underlying self-disclosure and multi-party information disclosure. In contrast to self-disclosure, multi-party privacy considerations may extend to encompass a wider interaction between individuals themselves and other co-owners [14,19]. Recognizing psychological ownership, individuals may consider not only their own privacy preferences but also those of other co-owners [19]. This mental process may also be shaped by their psychological social distance from other co-owners [93]. Hence, their sharing decisions can vary, depending on the extent to which they attribute ownership to themselves in relation to other (proximal versus distal) co-owners. Future research may be interested in understanding how psychological ownership and social distance can affect the disclosure of others' information and the disclosure of collectively owned information.

We propose the following research questions:

- P1: Why and how do people share private information about others?
- P2: Why and how do people share private information that is collectively owned with others?
- P3: What are the differences in the underlying mechanisms between the sharing of others' information and that of collectively owned information?

### *5.2. Theory: an individual-level cost-benefit trade-off and its multilevel extensions*

PID has been explained primarily via the calculus lens in IS research [3,50]. The development of a multilevel theoretical framework based on the calculus perspective is therefore essential. To do this, we start by discussing possible extensions of the calculus lens at the individual level, demonstrating how it can be extended to account for multi-party information disclosure (Section 5.2.1). Then we delineate its group-level extensions, discussing how group-level factors can influence the individual-level calculus and how social groups may leverage the calculus lens (Section 5.2.2). Finally, we elucidate how IT artifact characteristics can affect the way the calculus lens is used by individuals and social groups (Section 5.2.3).

We recognize the importance of other theories shown in Table 2. However, this study is more IS-focused and therefore aims to provide a theoretical framework to guide IS research. In this sense, the calculus lens is more pertinent to our study. The multilevel extensions of other theories in Table 2 surpass the scope of this paper. We acknowledge this limitation, discuss it in Section 6, and put forward opportunities for future research.

#### *5.2.1. Self-centric calculus versus others-centric calculus*

Past IS research has primarily deemed the calculus lens to be a self-

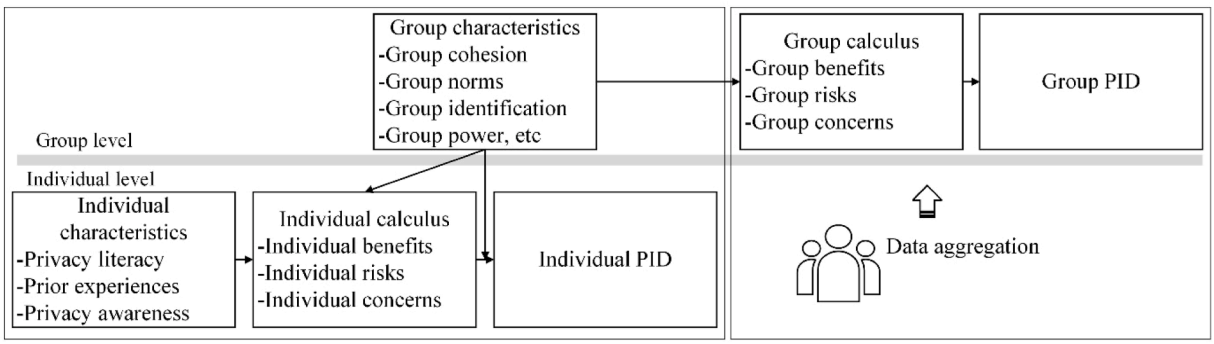



**Fig. 7.** Social calculus: two possible avenues.

centric mental process, in which individuals base their cost–benefit evaluations on personal interests [2]. Through this approach, information privacy is deemed a "commodity" that can be exchanged for personal gains. However, information privacy sometimes is also co-owned and co-regulated by multiple stakeholders [19]. In such situations, individuals' disclosure decisions can have potential ramifications for other people, thus requiring their privacy reflections upon all co-owners involved. Therefore, we propose that individuals may perform an others-centric cost–benefit assessment under conditions in which shared information involves multiple co-owners.

This idea can be supported theoretically by prior IS literature. For example, when privacy decisions have interpersonal impacts, individuals can project others' privacy risks [94], predict others' privacy concerns [40], or evaluate others' privacy benefits [95]. Recent research has also shown that others' benefits and risks can predict co-owned information disclosure [47]. Resorting to psychology theories, the above mental processes are inherently rooted in social cognition, an ability to interpret and understand other people and their social situations [96]. Indeed, human beings are able to think and act in interdependent ways [97,98]. Individuals can express empathic concern toward others in pain, manifest happiness toward those in joy, and more importantly, project and predict others' thoughts and beliefs [99]. In fact, numerous psychology theories have accounted for these mental processes, particularly documented by the theory of mind [100] and interpersonal reactivity index (e.g., empathy and perspective-taking; [97]).

Future research may be interested in exploring what drives self-centric calculus versus others-centric calculus. Drawing upon prior theories, here we offer several possible avenues (see Fig. 6). First, future research can consider interpersonal power, a potential inhibitor for perspective-taking and social cognition in interpersonal communication [101]. According to past research, individuals with more influence or control over other people may anchor heavily to their own viewpoints, thereby reducing their ability to project others' thoughts and ideas [101]. Second, it is also necessary to consider self-construal, a concept that captures how people perceive themselves in relation to others (i.e., interdependent self and independent self; [102]). Interdependent self-construal could increase social cognition [103]. Using this idea, we speculate that self-construal can also affect the extent to which individuals engage in self-centric calculus or others-centric calculus. Third, future research can also consider social value orientations, which capture how people tend to allocate resources to themselves against others [104]. Those who exhibit altruistic orientations are more inclined to consider others' welfare and behave in a prosocial way [104]. Using this idea, we can speculate that social value orientations can also affect others-centric calculus. Last, individuals with dark triad personality traits (e.g., narcissism and psychopathy) are often manipulative, lack empathy, and tend to prioritize their own interests [105]. Hence, we speculate that dark triad personality traits may influence the weight people give to their own calculus (e.g., self-benefits) against others' calculus.

All in all, we propose the following questions for future research to consider:

- P4: How do individuals use an others-centric calculus approach when they share multi-party information?
- P5: What are the factors that move privacy calculus from being self-centric to others-centric?
- P6: How do we compare the effects of the self-centric calculus and others-centric calculus?

#### 5.2.2. From individual calculus to social calculus

The calculus lens is primarily an individual-level approach. It relatively overlooks the broader social contexts within which the mental calculus occurs and the higher-level social dynamics that guide, shape, or even dominate it. There have been several studies suggesting that individual calculus should progress toward social calculus by incorporating group dynamics (e.g., group cohesion and group norms; [1,14]). Following this prior research, we propose that social calculus is not only an individual-level calculus with group-level extensions but also a mental calculus approach leveraged by social groups as a whole (Fig. 7).

First, according to the theory of multilevel information privacy management [14], higher-level factors represent both opportunities and constraints, so that individuals' PID decisions are subtly shaped by the groups within which they are nested. Drawing upon this notion, we put forth that individual calculus may be skewed by group-level factors, which in turn increases or decreases PID at the individual level. For example, group identification may reduce the weight people give to privacy concerns when sharing information in a cohesive and close-knit group, thereby fostering self-disclosure [68]. In addition, social groups often have a shared understanding of acceptable behavior, namely group norms, which can also affect how information should be managed by individuals in these groups [14]. Thus, individuals may use the mental calculus differently, depending on whether group norms support their PID. Furthermore, social groups are inherently structured in a hierarchical way, meaning that group power dynamics represent another important higher-level factor for individual PID [106]. For example, sharing a group photo with a manager in a work setting may raise different privacy considerations than sharing a group photo with peers. In a power-unbalanced situation, we speculate that individuals may assign more weight to the negative consequences of sharing decisions, inducing them to withhold information to avoid judgment from those in positions of power. Last, it is also worth exploring whether the influence of group-level factors depends on contextual factors such as data sensitivity and data ownership [14]. For example, when co-owned information is shared, group norms may exert a stronger influence, as perceived co-ownership may heighten the salience of group identity and collective expectations [14]. Future research can examine how such contextual factors may moderate group-level effects on individual calculus, thereby providing insights into potential interaction effects.

Hence, we propose the following questions as possible future directions:

- P7: What are the group-level factors that can influence individuals' mental calculus?
- P8: How do group-level factors influence individuals' mental calculus and consequently result in individual PID?
- P9: How do contextual factors influence the effects of group-level factors on individual calculus?

Second, according to Laufer and Wolfe [16] and Westin [17], information privacy is a multilevel concept, meaning that social groups as a whole can claim rights to and exert control over group privacy. Hence, we argue that the calculus lens can be adopted at the group level, so that groups as a whole may, for instance, express collective privacy concerns. In fact, there have been several studies that have started examining this idea [24,107,108].

Building upon the APCO framework (antecedents—privacy calculus—outcomes; [50]), we propose that group-level calculus relies on group-related characteristics such as group size and group centrality [1]. Group characteristics are intricately related to the core criteria that define a group [88]. A social group can be classified in different

**Table 5**
Future recommendations for multilevel data collection and analysis.

| Procedures | Recommendations | Explanations |
|---|---|---|
| Multilevel data sampling | Ensuring between-group variability and within-group agreement | • Having larger samples at higher levels than the lower levels<br>• Fifty groups as a recommended threshold<br>• A group size of five for family contexts and thirty for education and organization contexts |
| Data aggregation and the emergence of collective constructs | Using composition to generate shared constructs | • Using descriptive analysis of units at lower levels to represent the property of social units at higher levels<br>• Each unit at the lower level contributes equally to a certain property at the higher level |
| | Using compilation to generate configurational constructs | • Using non-linear analysis of units at lower levels to represent the property of social units at higher levels<br>• Each unit at the lower level contributes unequally to a certain property at the higher level |
| | Using objective statistics to represent global constructs | • Using descriptive characteristics of a group to represent global constructs<br>• Objective attributes of a group do not require data aggregation |
| Multilevel data analysis | Statistically justifying data aggregation, including inter-rater reliability, inter-rater agreement, and between-group variability | • Intraclass correlation coefficients (1)<br>• Intraclass correlation coefficients (2)<br>• Eta-squared<br>• Within and between analysis (WABA)<br>• $r^{WG}$<br>• Average deviation |
| | Testing cross-level effects, including cross-level direct, cross-level moderation, and cross-level mediation | • HLM<br>• MSEM<br>• MLwiN<br>• LIMDEP 8.0<br>• R programming<br>• Mplus<br>• SPSS (linear mixed effects model) |

categories, hence yielding various taxonomies [1]. The taxonomies include the following:

- By intragroup cohesion: primary groups (e.g., family), secondary groups (e.g., school and organization), collectives (e.g., tourist group), and categories (e.g., ethnicity, gender, and age group).
- By goals: economic group (e.g., a company), political group (e.g., a political party), religious group, educational group, etc.
- By cultural values: individualistic cultures, collectivistic cultures, etc.

As an illustration, we expand upon the ideas put forward by Belanger and Crossler [1] and exemplify how collective privacy concerns manifest differently within family, organizational, and cultural/societal groups.

- *Family groups* separate from other social groups by their inherent intragroup cohesion and emotional bonds [19]. The cohesion within a family unit often arises from feelings of kinship, loyalty, and mutual support, in contrast to secondary groups' intragroup cohesion. Thus, to understand family privacy concerns, it is crucial to prioritize the effects of within-group cohesion and emotional bonds among family members [19]. Additionally, family privacy concerns may arise to the level of the highest concerned member, be a function of the family leader's concerns, or be a function of all the members' concerns combined [1].
- *Economic groups*, for example a company, despite exhibiting a wide range of goals, distinguish themselves from other groups by their profit-driven nature. Consequently, organizational privacy concerns may be shaped by factors linked to their profit-driven aims, including customer attitudes and preferences, internal protocols and management processes, external regulations, and competitive dynamics [1]. Hence, Belanger and Crossler [1] propose that organizational privacy concerns typically reflect the overall concerns harbored by the managerial team, encompassing (1) the privacy of the organization's information and management processes, and (2) the privacy of the organization's customers. In this sense, organizational privacy concerns usually derive from group leaders, contrasting with family privacy concerns that may involve all family members.
- *A cultural or societal group* is defined by the cultural values (e.g., power distance and uncertainty avoidance) embraced by its members. Belanger and Crossler [1] suggested that societal privacy concerns refer to the overall concerns citizens harbor for the privacy of their information within a cultural or societal group. Societal privacy concerns differ from family privacy concerns and organizational privacy concerns because they are deeply interwoven with the values of a cultural or societal group, rather than specific to a family unit or an organization.

Taken together, we propose the following questions:

- P10: How do different types of social groups apply the calculus lens at the group level?
- P11**:** How does group calculus influence PID at the group level?

#### *5.2.3. Emerging IT artifacts and group privacy calculus*

IT entities afford various forms of control that individuals and social groups can utilize to make their PID decisions in a digital environment [24]. Accordingly, the characteristics of IT entities play an important

role in multilevel (i.e., both individuals and groups) privacy management [14]. In fact, many prior studies have examined this idea at the individual level, showing that IT artifact characteristics (e.g., privacy control and privacy settings) can shape individual PID decisions [109, 110].

Despite its valuable contributions, prior research has often overlooked how IT artifact characteristics can also shape the way social groups use the calculus lens and engage in collective privacy practices. This theoretical angle is needed for two reasons. First, some social groups are inherently nested within IT entities (e.g., a work team on Slack). This nested structure therefore renders not only each individual member but also the group itself subject to IT-induced influences. For example, when using external communication channels on social media, social activist groups may express concerns about leaking internal group information (e.g., group chat history) to outsiders [24]. Second, emerging technologies (e.g., big data analytics) transcend the individual privacy landscape and extend into the realm of group privacy [88,92]. With the ubiquitous monitoring and processing of user granular data, individuals can be categorized into different algorithm-determined groups based on their behavioral patterns [88]. Drawing inferences from group profiles, these insights can be further exploited to manipulate groups at a larger scale, increasing group vulnerability to privacy threats [88].

It is thus important to understand what emerging IT characteristics affect group privacy management (e.g., group PID), and how. This insight can contribute to group privacy protection. Here we exemplify two possible avenues: group privacy tools and group inference technologies. Prior research has called for the design of group privacy protection tools. Indeed, current privacy tools are mostly individual-oriented [24], and do not explicitly consider the nested structures of individuals within groups [108]. Future research can explore how to design group privacy tools and how such tools can reduce group privacy concerns and facilitate group privacy management. Furthermore, future research can consider group inference technologies and their impacts on group calculus. For example, some studies have started examining how individuals express privacy concerns toward the group to which they belong, when they engage on digital platforms embedded with group inference algorithms [111].

Taken together, we propose the following research topics:

- P12: What are the emerging IT characteristics that can influence the way social groups use the calculus lens and make PID decisions?
- P13: How do emerging IT characteristics influence the way social groups use the calculus lens and make PID decisions?

### 5.3. Methodology: multilevel data collection and analysis

#### 5.3.1. Multilevel data sampling

For multilevel research, because there are at least two levels of data units, each unit should contain sufficient data to minimize the risk of range restriction, a potential issue that may impact the generalizability of results [28]. Some suggest that the 30/30 rule of thumb is the minimum threshold to reach statistical power for cross-level analysis [27]. This enables each level to have sufficient samples to reach enough statistical power. In practice, the recommended threshold is 50 groups, with a group size of 30 for the education context and 5 for the family context [112]. Indeed, it is more important to have a larger number of groups than a large number of individuals per group, and 50 groups or fewer would usually result in biased estimates [112]. In essence, the key consideration for multilevel data sampling is to ensure within-group homogeneity and between-group variability [89].

For example, to explore the influence of cultural values on privacy concerns, researchers can begin by choosing countries as the higher-level samples, considering cultural dimensions such as power distance and individualism. They should aim to select countries representing a diverse range across these cultural dimensions to ensure significant between-country variations. Subsequently, within each chosen country, researchers can focus on populations with similar demographic traits such as age and education. This step ensures that samples within each country exhibit similarities at the within-country level.

#### 5.3.2. Multilevel data aggregation

Another challenging task for multilevel research is the emergence of collective constructs, namely data aggregation [28]. Researchers need to clarify why data collected at the individual level can be aggregated to group-level analysis (theoretically) and examine whether there is some agreement in individual assessment (statistically).

There are two data aggregation principles, composition and compilation, that can be used to generate *shared constructs* and *configurational constructs,* respectively [30,48].

- *Composition* refers to the process of using descriptive statistics such as means and standard deviations to associate lower-level data with higher-level constructs [48]. For example, to generate group privacy concerns, researchers can collect each group member's privacy concerns and use the average value to represent group-level privacy concerns. In this case, each group member contributes to group privacy concerns equally. Hence, the composition approach is suitable for the emergence of shared constructs, when group members have certain levels of shared understanding.
- *Compilation* is the process of combining data in a nonlinear manner, which is not reducible to all group members [48]. For example, within a group, individuals can vary in their privacy literacy and awareness. Because of these differences, when trying to assess the group's overall privacy concerns, it may be more accurate to focus on the views of those who are (1) more privacy-aware, (2) the most concerned about privacy, or (3) in leadership positions (since they may influence or represent the group's stance). Thus, the compilation approach is pertinent to the generation of configurational constructs, where group members may not equally contribute to a group-level construct.

#### 5.3.3. Multilevel data analysis

To inspire multilevel PID research, we provide a summary of multilevel analytical tools and procedures (see Table 5). There are multiple data analytical techniques pertinent to multilevel research, including hierarchical linear modeling (HLM) and multilevel structural equation modeling (MSEM). Using these data analytical techniques, researchers can calculate indices that capture within-group agreement and between-group variability, to justify data aggregation [30,89]. These indices include $r_{WG}$, average deviation, and intraclass correlation coefficients. In addition, researchers can use these techniques to analyze how variables at one level can influence variables at other levels (i.e., cross-level effects).

Overall, putting Section 5.3.1, Section 5.3.2, and Section 5.3.3 together, we propose the following research questions:

- P14: How do we use indices such as intraclass correlation coefficients to justify between-group variability and within-group agreement when conducting multilevel PID research?
- P15: How do we use multilevel data analytical tools to conduct multilevel PID research?

Taking these together, we have addressed RQ3 in Section 5.1, Section 5.2, and Section 5.3. We present a multilevel theoretical framework and propose avenues for future research, encouraging scholars to transition from individual level to multilevel investigation in this area. We highlight the need to focus on multi-party information disclosure and evolve from an individual-level to a multilevel perspective in privacy calculus theory. Furthermore, we outline key procedures, analytical methods, and tools relevant to multilevel PID research. This includes a synthesis of multilevel data sampling, data aggregation, and cross-level analysis. These insights can guide future research on this topic.

## 6. Discussion and conclusions

### 6.1. Theoretical contributions

Our research makes several notable theoretical contributions to the IS literature. First, we not only provide an up-to-date synthesis of PID research across diverse contexts but also uncover the foundational theoretical perspectives undertaken by the literature. These findings can advance IS research on factors, at different levels of theorizing and analysis, that influence PID decisions. Specifically, we examined the foundational theoretical perspectives toward information privacy undertaken by prior studies, finding that PID can be viewed as an outcome of (1) a cost–benefit mental assessment, (2) control over information flow in interpersonal communication, and (3) compliance with social norms rooted in individuals' most salient identity in a social context. These findings suggest the need for multilevel theorizing and considering factors across levels. Considering such factors and theories can significantly improve our understanding of PID and make our models more realistic and nuanced.

Second, we confirm the multilevel nature of PID. We put forth that (1) "private information" sometimes encompasses personal details concerning multiple stakeholders, and hence the sharing of such information extends beyond the individual level, and (2) individuals are often nested within a hierarchical system of human social units and IT units, and their PID decisions should be affected by such a hierarchical social structure. Thus, our synthesis provides an important first step toward multilevel research on PID, not only establishing its multilevel nature but also paving the way for future empirical research on it.

Third, we provide a multilevel theoretical framework together with a set of propositions for future multilevel PID research. Our proposed framework extends the current individual-level self-centric privacy calculus to a multilevel others-centric approach. Privacy calculus theory has been viewed as a self-centric approach that individuals adopt to deliberate on the benefits and risks to themselves when sharing their private information. We suggest that such a self-centric calculus approach should be extended to an others-centric perspective, meaning that individuals can also project the benefits and risks to others when shared information affects multiple co-owners. In addition, privacy calculus is an individual-level theory that lacks considerations of higher-level social units. We suggest that the individual calculus should progress toward a social calculus approach. This means that not only should the individual calculus consider its group-level extensions, but also social groups can adopt the calculus lens as a whole. The above multilevel extensions to the calculus lens can provide important theoretical foundations for future multilevel PID research.

Last, we summarize the main procedures, analytical approaches, and tools pertinent to multilevel PID research. We synthesize recommendations for multilevel data sampling and data aggregation, demonstrating how to ensure between-group variability and within-group agreement, as well as how to generate collective constructs (i.e., shared constructs and configurational constructs). Then, we also delineate some of the analytical tools that can be used to justify data aggregation and execute cross-level analysis, such as HLM and MSEM. All in all, the synthesis of multilevel analytical procedures can provide important guidance on how to conduct multilevel PID research.

### 6.2. Practical contributions

This paper has several practical implications for multi-party privacy protection. It points to the importance of mitigating privacy threats occurring to multiple stakeholders nested within groups. In a digital environment where information privacy is usually interdependent, it is increasingly important to achieve individual privacy through collective efforts [92]. This is because one might violate others' privacy by accidentally disseminating sensitive information about other people [40]. In addition, with the ubiquitous implementation of AI algorithms across various digital platforms, inferential privacy about users is based on not only what they self-disclose but also on what other people disclose about them [88]. Therefore, it is crucial to manage multi-party information, as its careless disclosure may bring multiplied privacy threats and expose all stakeholders to privacy risks [5].

However, the existing privacy-enhancing technologies, privacy settings, and protective tools are mostly individual-oriented, primarily focusing on the needs and preferences of individual users rather than those of groups [24]. Such tools, however, may not invariably be effective in an interdependent digital environment. Thus, we suggest that extant privacy settings and tools need to consider not only individuals' privacy preferences, but also the social contexts and the groups within which individuals are nested. To this end, practitioners can develop privacy tools, settings, and functions that facilitate privacy protection pertaining to multiple stakeholders and multilevel social units (e.g., dyads and family groups).

### 6.3. Limitations and opportunities for future research

This paper has some limitations that point to the need for additional research. First, it is inevitable that some papers have been excluded due to search query formulation, database selection, screening, and full article assessment. For example, we recognize that there are some AIS conference papers studying interdependent disclosure and peer disclosure [113] and disclosure of others' information [114–118]. However, such papers tend to emerge as journal papers after a more careful review process. We thus focused on these post-review papers. In addition, our focus was on quantitative behavioral studies on PID. This was done to limit the scope of the review and to synthesize papers from similar theoretical and research traditions. However, this resulted in excluding qualitative papers and other research genres. Hence, we advocate that future research explore not only the IS literature but also publications in relevant disciplines, as well as key conference papers (e.g., by AIS), to provide a more comprehensive review.

Second, during the coding process, we did not distinguish between independent variables, mediating variables, and moderating variables. We also did not synthesize past findings statistically, through meta-

analysis techniques. Future research can use a more nuanced coding approach to unpack the intricacies of antecedents and their specific effects on PID, such as using meta-analysis techniques.

Third, we presented two potential methods to expand the privacy calculus lens into a multilevel theory. However, we did not explore the adaptation of other theories (e.g., social exchange theory) into multilevel frameworks extensively. This is because we emphasized an IS-centered viewpoint. We encourage future studies to consider how other theories can be adapted to multilevel ideas. Future research can also consider how to use qualitative methods for multilevel investigation, for example, interviewing family groups and examining how family group dynamics influence each family member's PID online.

Last, while we have highlighted several significant propositions for future studies, it is important to acknowledge that there may be additional avenues for extension that we have not explicitly delineated. For example, existing IS research predominantly utilized cross-sectional methodologies, with limited longitudinal perspectives. However, the dynamics of decision-making processes and the cognitive mechanisms driving PID may be subject to temporal factors [14,19]. Future research should consider such extensions and not limit itself to our propositions. For example, it can consider longitudinal perspectives on PID and move from intentions to actual PID behavior.

## Author statement

We, the co-authors of the manuscript (ID: INFMAN-D-23-01047R3) hereby confirm that all of us have made significant contributions to this paper, and all authors have approved the current version of the manuscript and agree to its re-submission to *Information & Management.*

## CRediT authorship contribution statement

**Mingxin Zhang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Reeva Lederman:** Writing – review & editing, Supervision, Conceptualization. **Ofir Turel:** Writing – review & editing, Supervision, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Research trend and study contexts

In Appendix A, we describe PID's research trend (from 2002 to 2024) and study contexts.
Fig. A1

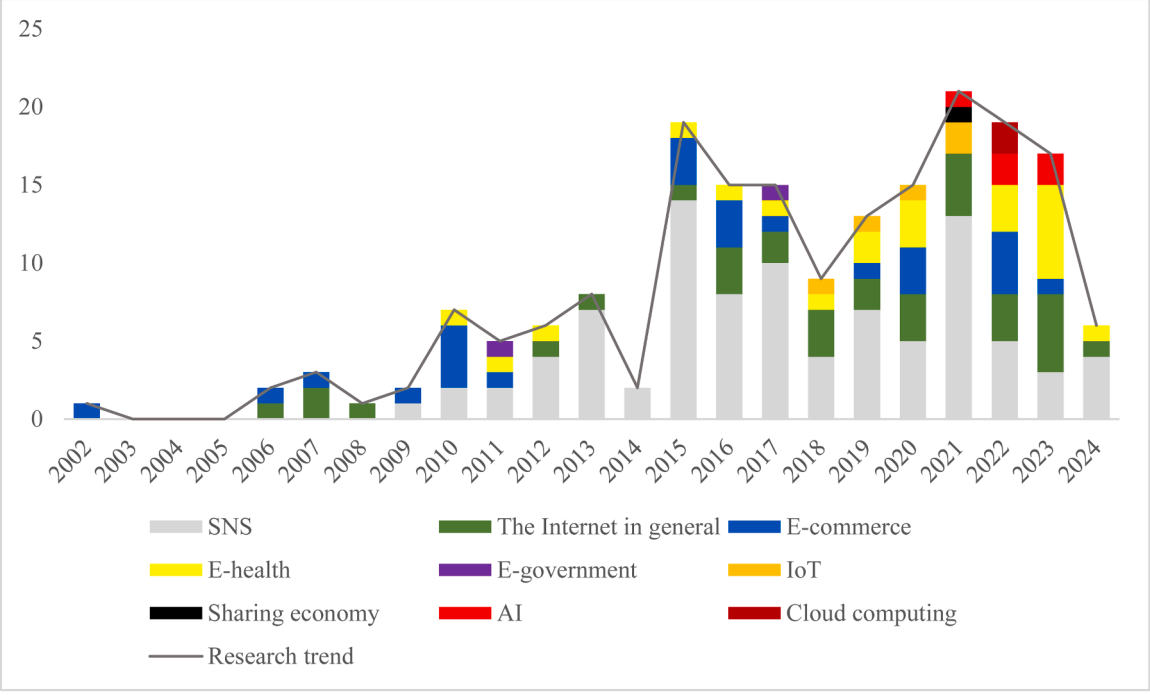



**Fig. A1.** PID's research trend and study contexts.

## Appendix B. The coded papers by publication outlets

In Appendix B, we summarize the coded papers by their publication outlets.
Table B1

**Table B1**
The coded papers and their publication outlets.

| No. | Publication Title | Database search or reference analysis | AIS Senior Scholar's Premium Journals | ABS journal ranking list (2021) | ABDC journal ranking list (2022) | Number of publications |
|---|---|---|---|---|---|---|
| 1 | *AIS Transactions on Human-Computer Interaction* | Database search | | 2 | A | 0 |
| 2 | *Australasian Journal of Information Systems* | Database search | | 2 | A | 0 |
| 3 | *Behaviour & Information Technology* | Database search | | 2 | A | 13 |
| 4 | *Communications of the ACM* | Database search | | 2 | A | 0 |
| 5 | *Communications of the Association for Information Systems* | Database search | | 2 | A | 3 |
| 6 | *Computers & Security* | Reference analysis | | None | A | 4 |
| 7 | *Computers in Human Behavior* | Database search | | 2 | A | 43 |
| 8 | *Decision Support Systems* | Database search | √ | 3 | A* | 8 |
| 9 | *Electronic Commerce Research* | Database search | | 2 | A | 1 |
| 10 | *Electronic Markets* | Reference analysis | | 2 | A | 2 |
| 11 | *Enterprise Information Systems* | Database search | | 2 | A | 0 |
| 12 | *European Journal of Information Systems* | Database search | √ | 4 | A* | 5 |
| 13 | *Government Information Quarterly* | Database search | | 3 | A | 1 |
| 14 | *Industrial Management & Data Systems* | Database search | | 2 | A | 2 |
| 15 | *Information & Management* | Database search | √ | 3 | A* | 9 |
| 16 | *Information, Communication & Society* | Reference analysis | | 2 | A | 2 |
| 17 | *Information and Organization* | Database search | √ | 3 | A* | 0 |
| 18 | *Information Society* | Database search | | 3 | A | 0 |
| 19 | *Information Systems Frontiers* | Database search | | 3 | A | 2 |
| 20 | *Information Systems Journal* | Database search | √ | 4 | A* | 3 |
| 21 | *Information Systems Research* | Database search | √ | 4* | A* | 7 |
| 22 | *Information Technology & People* | Database search | | 3 | A | 11 |
| 23 | *International Journal of Electronic Commerce* | Database search | | 3 | A | 3 |
| 24 | *International Journal of Information Management* | Database search | | 2 | A* | 11 |
| 25 | *International Journal of Medical Informatics* | Reference analysis | | None | A | 3 |
| 26 | *Internet Research* | Database search | | 3 | A | 7 |
| 27 | *Journal of Advertising Research* | Reference analysis | | 3 | A | 1 |
| 28 | *Journal of Computer-Mediated Communication* | Database search | | 3 | A | 6 |
| 29 | *Journal of Computer Information Systems* | Database search | | 2 | A | 1 |
| 30 | *Journal of Consumer Behaviour* | Reference analysis | | 2 | A | 3 |
| 31 | *Journal of Consumer Psychology* | Reference analysis | | 4* | A* | 2 |
| 32 | *Journal of Enterprise Information Management* | Database search | | 2 | A | 2 |
| 33 | *Journal of Global Information Management* | Database search | | 2 | A | 2 |
| 34 | *Journal of Information Technology* | Database search | √ | 4 | A* | 1 |
| 35 | *Journal of Management Information Systems* | Database search | √ | 4 | A* | 4 |
| 36 | *Journal of Strategic Information Systems* | Database search | √ | 4 | A* | 4 |
| 37 | *Journal of the Association for Information Science and Technology* | Database search | | 3 | A* | 6 |
| 38 | *Journal of the Association for Information Systems* | Database search | √ | 4* | A* | 4 |
| 39 | *MIS Quarterly* | Database search | √ | 4* | A* | 5 |
| 40 | *Scandinavian Journal of Information Systems* | Database search | | 2 | A | 0 |
| 41 | *Technological Forecasting and Social Change* | Reference analysis | | 3 | A | 1 |
| 42 | *The Database for Advances in Information Systems* | Database search | | 2 | A | 4 |

## Appendix C. The coded papers by publication year and publication outlet

In Appendix C, we summarize the coded papers by publication year and publication outlet.
Table C1

**Table C1**
The coded papers by publication year and publication outlet.

| | 2002 | 2003 | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *BIT* | | | | | | | | | | | | | | 1 | | 4 | 1 | | | 1 | 2 | 2 | 2 |
| *CAIS* | | | | | | | | | | | | | | | | 1 | | 1 | | | | 1 | |
| *C&S* | | | | | | | | | | | | | | | | | | | 2 | 1 | | 1 | |
| *CHB* | | | | | | 1 | | | | 2 | 3 | 3 | 1 | 7 | 3 | 4 | 3 | 1 | 3 | 3 | 2 | 2 | 1 |
| *DSS* | | | | | | | | | 2 | 1 | | 1 | | 1 | | | 1 | | | | 1 | 1 | |
| *ECR* | | | | | | | | | | | | 1 | | | | | | | | | | | |
| *EM* | | | | | | | | | | | | | | 1 | | | | | 1 | | | | |
| *EJIS* | | | | | | | | | 1 | | | | | 2 | | 2 | | | | | | | |
| *GIQ* | | | | | | | | | | | | | | | | | | | 1 | | | | |
| *IMDS* | | | | | | | | | | | | | | | | | | 1 | | | 1 | | |
| *IM* | | | | | | 1 | | | 1 | | | | | | 2 | 1 | 1 | | | 2 | | | |
| *ICS* | | | | | | | | | | | | | | | | | | | | | | 2 | |
| *ISF* | | | | | | | | | | | | 1 | | | | | | 1 | | | | | |
| *ISJ* | | | | | | | | | | | | | | 1 | | | | 2 | | | | 2 | |
| *ISR* | | | | | 1 | | | | | 1 | | 1 | | | | | | 1 | | | 1 | 2 | |
| *ITP* | | | | | | | | | | 1 | | | | 1 | | | | 1 | 1 | 2 | 2 | 1 | |
| *IJEC* | | | | | | | | | 1 | | 1 | | | | | | | 1 | | | | | |
| *IJIM* | | | | | | | | 1 | | | | | | | 2 | 1 | | | 2 | 2 | 1 | | |
| *IJMI* | | | | | | | | | | | | | | | | | | 1 | | | | 2 | |
| *IR* | | | | | | | | | | | | | | 1 | 1 | | | | 1 | 1 | 1 | | 1 |
| *JAR* | | | | | | | | | | | | | | | | | | | | | 1 | | |
| *JCMC* | | | | | | | | | | | | | 1 | | 1 | 1 | 1 | | 1 | | | | |
| *JCIS* | | | | | | | | | 1 | | | | | | | | | | | | | | |
| *JCB* | | | | | | | | | | | | | | | 1 | | | | | | 1 | 1 | |
| *JCP* | | | | | | | | | | | | | | | | | | | | 1 | 1 | | |
| *JEIM* | | | | | | | | | | | | | | | | | | 1 | | 1 | | | |
| *JGIM* | | | | | | | | | | | | | | | | | | | | 1 | | | |
| *JIT* | | | | | | | | | 1 | | | | | | | | | | | 1 | | | |
| *JMIS* | | | | | | | | 1 | | | | | | 1 | | 1 | | | | 1 | | | |
| *JSIS* | 1 | | | | | | 1 | | | | | 1 | | 1 | | | | | | | | | |
| *JASIST* | | | | | | | | | | | | | | 1 | 1 | | | 1 | | | 1 | | |
| *JAIS* | | | | | | | | | | | | | | | | 1 | | 1 | | | | | 1 |
| *MISQ* | | | | | 1 | 1 | | | | | | | | | | | | | | 1 | 1 | 1 | |
| *TFSC* | | | | | | | | | | | | | | | | | | | | | 1 | | |
| *TDAIS* | | | | | | | | | | | | | | | | | | | 1 | 1 | 1 | | |

## Appendix D. The complete list of the coded papers ($N = 186$)

In Appendix D, we present the complete list of coded papers.
Table D1

**Table D1**
The complete list of coded papers.

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Self-disclosure | Likert scale | Self | 1043 respondents from Germany | Longitudinal surveys | Individual | Individual-level | • Negative experience<br>• Privacy concerns<br>• Data protection<br>• Gender | None | The Internet in general | None | *Behaviour & Information Technology* |
| 2 | Self-disclosure | Likert scale | Self | 1500 Canadian adults | Cross-sectional surveys | Individual | Individual-level | • Gratification<br>• Trust in SNS<br>• Trust in SNS users<br>• Concerns about social threats<br>• Concerns about organizational threats<br>• Social media literacy<br>• Individual privacy management | Privacy calculus theory<br>Use and gratification theory | SNS | Social media app | *Behaviour & Information Technology* |
| 3 | Self's sharing of information ownership | Likert scale | Others | 400 Facebook college students in the US | Cross-sectional surveys | Individual | Individual-level | • Others' sharing of information ownership<br>• Self's privacy concerns as a co-owner<br>• Positive norms of shared information ownership<br>• Privacy rule adaptation | Communication privacy management theory | SNS | Social media app | *Behaviour & Information Technology* |
| 4 | Intention to disclose personal information | Likert scale | Self | 807 participants in the US | Cross-sectional surveys | Individual | Individual-level | • Perceived privacy risks<br>• Perceived value of data transfer to the cloud<br>• Dispositional traits<br>• User experience | Privacy calculus theory | Cloud computing | Mobile cloud computing app | *Behaviour & Information Technology* |
| 5 | Disclosure of personal information | Likert scale | Self | 616 participants in the US | Experiment | Individual | Individual-level | • Privacy risk beliefs<br>• Privacy concerns<br>• Perceived information control<br>• Self-presentation motivation | Privacy calculus theory | The Internet in general | None | *Behaviour & Information Technology* |
| 6 | Urge to share status | Likert scale | Self | 523 microblogging users in China | Cross-sectional surveys | Individual | Individual-level | • Impulsiveness<br>• Inherent novelty seeking<br>• Positive emotion<br>• Negative emotion | None | SNS | Microblog | *Behaviour & Information Technology* |
| 7 | Information sharing performance | Checklist | Self | 80 dyads of couples from China and Germany | Experiment | Dyad | Individual-level | • Interpersonal trust<br>• Self-construal<br>• Cultural orientation | Self-construal theory | The Internet in general | None | *Behaviour & Information Technology* |
| 8 | Emotional disclosure | Likert scale | Self | 293 student Facebook users in the US | Cross-sectional surveys | Individual | Individual-level | • Big Five personality<br>• social support | None | SNS | Social media app | *Behaviour & Information Technology* |
| 9 | Intention to disclose information | Likert scale | Self | 457 social networking website users | Cross-sectional surveys | individual | Individual-level | • Prior positive experiences<br>• Trust in website<br>• Privacy concerns | Privacy calculus theory<br>Prospect theory | SNS | Social media app | *Behaviour & Information Technology* |
| 10 | The disclosure of personal data | Likert scale | Self | 410 adult Portuguese citizens | Cross-sectional surveys | Individual | Individual-level | • Trust in public institutions<br>• Privacy concerns regarding public institutions | None | E-government | None | *Behaviour & Information Technology* |
| 11 | Sharing of information | Checklist | Self | 125 university students | Experiment | Individual | Individual-level | • Social motivation<br>• Time pressure<br>• Information display | None | The Internet in general | None | *Behaviour & Information Technology* |

(*continued on next page*)

**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 12 | Self-disclosure | Likert scale | Self | 506 participants | Cross-sectional surveys | Individual | Individual-level | • Privacy-related nudges<br>• Perceived control<br>• Perceived privacy risks<br>• Trust in provider | Privacy calculus theory | SNS | Social media app | *Behaviour & Information Technology* |
| 13 | Information disclosure | Likert scale | Self | 1159 participants in the US, Korea, and China | Cross-sectional surveys | Individual | Individual-level | • Information privacy concerns<br>• Interactional privacy concerns<br>• Psychological privacy concerns<br>• Perceived risks<br>• Contextual factors: Offline connection, mutual friends, same city, same college, and same SNS group | Contextual integrity | SNS | Social media app | *Behaviour & Information Technology* |
| 14 | Willingness to disclose health information | Likert scale | Self | 519 participants | Cross-sectional surveys | Individual | Individual-level | • Perceived benefits<br>• Perceived transparency of privacy statement<br>• Familiarity with health information exchange<br>• Cognitive trust<br>• Emotional trust | Theory of reasoned action | E-health | HIE technology | *Communications of the Association for Information Systems* |
| 15 | Willingness to share personal health information | Likert scale | Self | 253 participants | Cross-sectional surveys | Individual | Individual-level | • Perceived threat severity<br>• Perceived threat susceptibility<br>• Perceived barriers<br>• Self-efficacy<br>• Perceived benefits<br>• Design cues | Health belief model | E-health | AI-driven health platforms | *Communications of the Association for Information Systems* |
| 16 | Self-disclosure | Likert scale | Self | 259 participants | Cross-sectional surveys | Individual | Individual-level | • Familiarity with big data<br>• Perceived control<br>• Perceived vulnerability<br>• Self-efficacy<br>• Privacy concerns<br>• Trust | Privacy calculus theory<br>Protection motivation theory<br>Theory of planned behavior | SNS | Social media app | *Communications of the Association for Information Systems* |
| 17 | Information disclosure intention | Likert scale | Self | 533 participants | Cross-sectional surveys | Individual | Individual-level | • Cognitive trust<br>• Emotional trust<br>• Perceived transparency of privacy statement | Theory of reasoned action | E-health | HIE technology | *Computers & Security* |
| 18 | Disclosure behaviour | Likert scale | Self | 318 IoT device owners | Experiment | Individual | Individual-level | • Scope of sharing<br>• Disclosure visualization<br>• Privacy concerns<br>• Trusting beliefs<br>• Risk beliefs | Privacy calculus theory | IoT | Smart home IoT devices | *Computers & Security* |
| 19 | Disclosure intention | Likert scale | Self | 465 participants in Malaysia | Cross-sectional surveys | Individual | Individual-level | • Information type<br>• Privacy concerns | None | The Internet in general | None | *Computers & Security* |
| 20 | Self-disclosure | Likert scale | Self | 720 participants in the US and UK | Cross-sectional surveys | Individual | Individual-level | • Perceived benefits<br>• Perceived risks<br>• Trust in provider<br>• Fear of missing out | Privacy calculus theory | SNS | Social media app | *Computers & Security* |
| 21 | Self-disclosure | Likert scale | Self | 315 undergraduate Facebook users | Cross-sectional surveys | Individual | Individual-level | • Ownership rule<br>• Permeability rule<br>• Linkage rule | Communication privacy management theory<br>Social penetration theory | SNS | Social media app | *Computers in Human Behavior* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 22 | Disclosure of information about others | Likert scale | Others | 364 undergraduate students in the US | Cross-sectional surveys | Individual | Individual-level | • Shallow disclosure<br>• Deep disclosure<br>• Social identity<br>• Big Five personality<br>• Prior privacy invasion<br>• Concerns about others' privacy<br>• Social norms<br>• Attitude toward using SNS | Theory of planned behavior<br>Theory of reasoned action | SNS | Social media app | *Computers in Human Behavior* |
| 23 | Willingness to share personal information | Coding | Self | 1941 participants | Experiment | Individual | Individual-level | • Regulatory focus<br>• Default settings<br>• Consensus | Regulatory Focus Theory | The Internet in general | None | *Computers in Human Behavior* |
| 24 | Self-disclosure | Checklist | Self | 222 participants in the US | Cross-sectional surveys | Individual | Individual-level | • Privacy concerns<br>• Social rewards<br>• Distant-future intentions<br>• Near-future intentions | Privacy calculus theory<br>Construal-level theory | SNS | Social media app | *Computers in Human Behavior* |
| 25 | Willingness to disclose personal information | Likert scale | Self | 949 participants | Cross-sectional surveys<br>Experiments | Individual | Individual-level | • Gamification<br>• Privacy benefits<br>• Privacy risks<br>• Privacy concerns<br>• Cognitive absorption | Privacy calculus theory | The Internet in general | Gamified app | *Computers in Human Behavior* |
| 26 | Disclosure behaviour | Checklist | Self | 122 undergraduate students in the US | Cross-sectional surveys | Individual | Individual-level | • Privacy attitudes<br>• Privacy policy consumption<br>• Privacy settings | None | SNS | Social media app | *Computers in Human Behavior* |
| 27 | Self-disclosure | Checklist | Self | 622 teens in the US | Cross-sectional surveys | Individual | Individual-level | • Demographics<br>• SNS use<br>• Friend type<br>• Trust<br>• Privacy settings | None | SNS | Social media app | *Computers in Human Behavior* |
| 28 | Information disclosure | Likert scale | Self | 746 participants in Singapore | Cross-sectional surveys | Individual | Individual-level | • Privacy concerns<br>• Parental mediation | None | The Internet in general | None | *Computers in Human Behavior* |
| 29 | Self-disclosure | Likert scale | Self | 1314 secondary education students | Cross-sectional surveys | Individual | Individual-level | • Attitude<br>• SN friends<br>• SN parents<br>• SN teachers<br>• Prototype favorability<br>• Prototype similarity | Prototype willingness model | SNS | Social media app | *Computers in Human Behavior* |
| 30 | Self-disclosure | Likert scale | Self | 661 Facebook users in Slovenia | Cross-sectional surveys | Individual | Individual-level | • Privacy awareness<br>• Privacy social norms<br>• Privacy policy<br>• Privacy control<br>• Privacy value<br>• Privacy concerns | Communication privacy management theory | SNS | Social media app | *Computers in Human Behavior* |
| 31 | Self-disclosure | Checklist | Self | 128 Internet users in Germany | Experiment | Individual | Individual-level | • Privacy concerns<br>• Social support<br>• Quantity feature<br>• Similarity feature | Privacy calculus theory<br>Technology affordance theory | SNS | Social media app | *Computers in Human Behavior* |
| 32 | Self-disclosure | Likert scale | Self | 322 WeChat users in China | Interviews<br>Cross-sectional surveys | Individual | Individual-level | • Perceived ephemerality<br>• Perceived usefulness<br>• Perceived controllability<br>• Perceived severity<br>• Perceived intrusion | Privacy calculus theory | SNS | Social media app | *Computers in Human Behavior* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 33 | Intention to share personal data | Likert scale | Self | 222 participants | Experiment | Individual | Individual-level | • Perceived<br>• Severity<br>• Response-efficacy<br>• Self-efficacy<br>• Response cost<br>• Situational severity<br>• Surveillance | Protection motivation theory | The Internet in general | None | *Computers in Human Behavior* |
| 34 | Self-disclosure | Likert scale | Self | 463 undergraduate students in the US | Cross-sectional surveys | Individual | Individual-level | • Self-construal<br>• SNS usage<br>• Personality traits | Self-construal theory | SNS | Social media app | *Computers in Human Behavior* |
| 35 | Willingness to provide personal information | Likert scale | Self | 525 participants | Cross-sectional surveys | Individual | Individual-level | • Innovativeness<br>• General privacy concerns<br>• Perceived usefulness<br>• Perceived convenience<br>• Perceived vulnerability<br>• Perceived severity<br>• Coping efficacy<br>• Self-efficacy | Privacy calculus theory<br>Protection motivation theory | SNS | Recommendation systems | *Computers in Human Behavior* |
| 36 | Self-disclosure | Likert scale | Self | 301 Facebook users | Cross-sectional surveys | Individual | Individual-level | • Demographics<br>• Big Five personality<br>• Social cohesion<br>• Virtual community<br>• Companionship<br>• Exhibitionism<br>• Relationship Maintenance<br>• Passing Time | Uses and gratifications theory | SNS | Social media app | *Computers in Human Behavior* |
| 37 | Self-disclosure | Checklist | Self | 451 participants in the US | Cross-sectional surveys | Individual | Individual-level | • Privacy concerns<br>• Big Five personality<br>• Trust<br>• Perceived information sensitivity<br>• Self-quantification | None | IoT | Wearable technology | *Computers in Human Behavior* |
| 38 | Self-disclosure | Likert scale | Self | 566 participants in Germany | Longitudinal surveys | Individual | Individual-level | • SNS use<br>• Social capital | Social capital theory | SNS | Social media app | *Computers in Human Behavior* |
| 39 | Information disclosure | Checklist | Self | 969 participants in the UK | Experiment | Individual | Individual-level | • Ego depletion<br>• Cognitive load<br>• Affect | None | The Internet in general | None | *Computers in Human Behavior* |
| 40 | Willingness to provide personal information | Likert scale | Self | 154 participants in the US | Cross-sectional surveys | Individual | Individual-level | • Perceived benefits<br>• Perceived risks<br>• Information sensitivity<br>• Trust<br>• Perceived critical mass<br>• Perceived compatibility<br>• Perceived complementarity | Privacy calculus theory | IoT | IoT devices in smart home, healthcare, and smart transportation | *Computers in Human Behavior* |
| 41 | SNS disclosure | Likert scale | Self | 527 adolescents in The Netherlands | Cross-sectional surveys<br>Experiment | Individual | Individual-level | • SNS use<br>• Social power<br>• in | None | SNS | Social media app | *Computers in Human Behavior* |
| 42 | Self-disclosure | Likert scale | Self | 718 SSN users | Cross-sectional surveys | Individual | Individual-level | • General trust<br>• Privacy concerns<br>• Perceived control over information | None | SNS | Social media app | *Computers in Human Behavior* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 43 | Intention to disclose information | Likert scale | Self | 429 participants in China | Cross-sectional surveys | Individual | Individual-level | • Privacy concerns<br>• Social presence of the interface<br>• Social presence of the interaction<br>• Social validation<br>• trust | Social presence theory<br>Technology acceptance model | E-health | Medical mobile app | *Computers in Human Behavior* |
| 44 | Social media distress disclosure | Likert scale | Self | 967 social media users in China | Cross-sectional surveys | Individual | Individual-level | • Perceived risks<br>• Perceived benefits<br>• Distressor | Privacy calculus theory | SNS | Social media app | *Computers in Human Behavior* |
| 45 | Personal information disclosure | Likert | Self | 246 participants | Cross-sectional surveys | Individual | Individual-level | • Persuasion knowledge<br>• Perceived benefits<br>• Perceived harms<br>• Attitude | None | E-commerce | Location-based advertising technology | *Computers in Human Behavior* |
| 46 | Private information disclosure | Coding | Self | 488 participants | Cross-sectional surveys | Individual | Individual-level | • Mobile phone usage<br>• Extent of OSN<br>• Application usability<br>• Perceived benefits<br>• Gender | None | SNS | Social media app | *Computers in Human Behavior* |
| 47 | Distress disclosure | Likert scale | Self | 398 college students | Cross-sectional surveys | Individual | Individual-level | • Network accessibility<br>• Visibility<br>• Visibility control<br>• Support-seeking goals<br>• Emotion-expressing goals<br>• Anticipated negative evaluation | None | SNS | Social media app | *Computers in Human Behavior* |
| 48 | Intention to disclose the location | Likert scale | Self | 164 undergraduate students in China | Cross-sectional surveys | Individual | Individual-level | • Utilitarian benefits<br>• Hedonic benefits<br>• Perceived benefits<br>• Perceived risks<br>• Perceived ease of use<br>• Services | Privacy calculus theory<br>Self-determination theory<br>Social role theory | SNS | Social media app | *Computers in Human Behavior* |
| 49 | Intention to share personal health information | Likert scale | Self | 180 participants in the US | Experiment | Individual | Individual-level | • Empowerment<br>• The use of Blockchain-based information systems | Privacy calculus theory | E-health | Blockchain-based information sharing systems | *Computers in Human Behavior* |
| 50 | Self-disclosure | Likert scale | Self | 127 undergraduate students in the US | Cross-sectional surveys | Individual | Individual-level | • Motives of Facebook use<br>• Relationship maintenance<br>• Passing time<br>• Entertainment<br>• Gender<br>• Privacy settings | None | SNS | Social media app | *Computers in Human Behavior* |
| 51 | Willingness to share information | Coding | Self | 173 MBA students in the US | Experiment | Individual | Individual-level | • Information ownership | None | The Internet in general | None | *Computers in Human Behavior* |
| 52 | Disclosure intention | Likert scale | Self | 273 SNS users in China | Cross-sectional surveys | Individual | Individual-level | • Perceived usefulness<br>• Perceived risk<br>• Subjective norm | Theory of planned behavior<br>Privacy calculus theory<br>Incentive theory | SNS | Social media app | *Computers in Human Behavior* |
| 53 | Decisions about sharing private data | Coding | Self | 1493 participants | Experiment | Individual | Individual-level | • Choice architecture including salience, effort, framing | Nudge theory | The Internet in general | Website | *Computers in Human Behavior* |
| 54 | Truthful disclosure | Likert scale | Self | 134 undergraduate students in the US | Cross-sectional surveys | Individual | Individual-level | • Behavioral activation systems<br>• E-health website evaluation<br>• Prevention regulatory focus<br>• Information sensitivity | Communication privacy management theory<br>Regulatory focus theory | E-health | Website | *Computers in Human Behavior* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 55 | Picture sharing preference | Likert scale | Co-owned | 29 participants in South Korea | Experiment | Individual | Individual-level | • Relationship closeness<br>• Ownership | Social exchange theory | SNS | Social media app | *Computers in Human Behavior* |
| 56 | Disclosure of personal information | Likert scale | Self | 236 undergraduate students in Canada | Experiment | Individual | Individual-level | • Information priming<br>• Gender | None | SNS | Social media app | *Computers in Human Behavior* |
| 57 | Disclosure of personally identifiable information | Coding | Self | 2253 participants in the US | Cross-sectional surveys | Individual | Individual-level | • Trust<br>• Privacy concerns | None | The Internet in general | None | *Computers in Human Behavior* |
| 58 | Personal information disclosure | Likert scale | Self | 780 secondary school students in Singapore | Cross-sectional surveys | Individual | Individual-level | • Narcissism<br>• Parental mediation | None | SNS | Social media app | *Computers in Human Behavior* |
| 59 | Self-disclosure | Checklist | Self | 100 Facebook users | Cross-sectional surveys | Individual | Individual-level | • Impulsivity<br>• Risk seeking<br>• Self-centeredness | Self-control theory | SNS | Social media app | *Computers in Human Behavior* |
| 60 | Disclosing personal data | Likert scale | Self | 208 Internet users in Dutch | Cross-sectional surveys | Individual | Individual-level | • Trust<br>• Perceived risks | None | E-government | Online transaction systems | *Computers in Human Behavior* |
| 61 | Disclosure of personal information | Checklist | Self | 328 college students in the US | Experiment | Individual | Individual-level | • Visual cues<br>• Linguistic cues<br>• Question type | None | E-health | Website | *Computers in Human Behavior* |
| 62 | Self-disclosure | Likert scale | Self | 64 participants in China | Experiment | Individual | Individual-level | • Anthropomorphism<br>• Social role<br>• Emotional attachment<br>• Satisfaction<br>• Social presence<br>• Privacy concern | Social role theory<br>Social presence theory | AI | Intelligent agent | *Computers in Human Behavior* |
| 63 | Self-disclosure | Likert scale | Self | 157 Facebook users in Australia | Cross-sectional surveys | Individual | Individual-level | • SNS use<br>• Stress<br>• Anxiety<br>• Impulsivity<br>• Depression<br>• Alexithymia<br>• Demographics<br>• Alcohol use | None | SNS | Social media app | *Computers in Human Behavior* |
| 64 | Willingness to disclose information | Likert scale | Self | 189 participants in the US | Cross-sectional surveys | Individual | Individual-level | • Perceived benefits<br>• Privacy concerns<br>• Data sharing expectations | Privacy calculus theory<br>Contextual integrity theory | IoT | Fitness technology | *The DATA BASE for Advances in Information Systems* |
| 65 | Information disclosure | Likert scale | Self | 242 participants in the US | Cross-sectional surveys | Individual | Individual-level | • Attentional impulsivity<br>• Motor impulsivity<br>• Non-planning impulsivity<br>• Privacy concerns | None | The Internet in general | None | *The DATA BASE for Advances in Information Systems* |
| 66 | Willingness to provide information | Likert scale | Self | 260 students in the US | Cross-sectional surveys | Individual | Individual-level | • Attitude<br>• Subjective norms<br>• Perceived behavioral control<br>• Perceived usefulness<br>• Ethical concerns<br>• Personal normative beliefs<br>• Perceived importance<br>• Anticipated regret | Theory of planned behavior | The Internet in general | None | *The DATA BASE for Advances in Information Systems* |
| 67 | Disclosure of Personal Health Information | Likert scale | Self | 529 participants | Experiment | Individual | Individual-level | • Argument Strength<br>• Attribute framing | Privacy calculus theory | E-health | Wearable health technology | *The DATA BASE for Advances in Information Systems* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 68 | Location disclosure | Likert scale | Self | 303 undergraduate students in the US | Cross-sectional surveys | Individual | Individual-level | • Social norms<br>• Attitude<br>• Perceived risks<br>• Perceived benefits<br>• Opinion leadership<br>• Privacy concerns<br>• Incentives<br>• Facilitating conditions | The Unified Theory of Acceptance and Use of Technology<br>Privacy calculus theory | SNS | Location-based social network applications | *Decision Support Systems* |
| 69 | Self-disclosure | Likert scale | Self | 486 students in China | Experiment | Individual | Individual-level | • Perceived justice<br>• Perceived privacy<br>• Prior invasion experience<br>• Privacy feedback<br>• Privacy choice | Justice theory<br>Contextual integrity theory | E-commerce | Mobile shopping app | *Decision Support Systems* |
| 70 | Information disclosure intentions | Likert scale | Self | 809 participants in China | Cross-sectional surveys | Individual | Individual-level | • Privacy disclosure experiences<br>• Privacy invasion experiences<br>• Privacy risks<br>• Privacy benefits | Heuristic-systematic model<br>Privacy calculus theory | The Internet in general | None | *Decision Support Systems* |
| 71 | Intention to disclose health information | Likert scale | Self | 367 students in the US | Cross-sectional surveys | Individual | Individual-level | • Health status<br>• Information sensitivity<br>• Big Five personality<br>• Privacy concerns<br>• Trust<br>• Previous invasion experiences<br>• Risk beliefs | None | E-health | website | *Decision Support Systems* |
| 72 | Disclosure intention | Likert scale | Self | 296 participants | Experiment | Individual | Individual-level | • Privacy risks<br>• Extroversion<br>• Frequency<br>• Receivers<br>• Location relevance<br>• Conditional value | Privacy calculus theory | The Internet in general | Location-based service technology | *Decision Support Systems* |
| 73 | Willingness to provide personal information | Likert scale | Self | 220 students in the US | Experiment | Individual | Individual-level | • Joy<br>• Fear<br>• Perceived relevance of information<br>• Awareness of privacy statement<br>• Information sensitivity<br>• Privacy risk beliefs | Stimulus–organism–response model<br>Privacy calculus theory | E-commerce | Website | *Decision Support Systems* |
| 74 | Information disclosure | Checklist | Self | 236 participants | Experiment | Individual | Individual-level | • Trust<br>• Privacy concerns<br>• Dyadic condition<br>• Privacy benefits | Privacy calculus theory<br>Theory of reasoned action<br>Social response theory | E-commerce | Website | *Decision Support Systems* |
| 75 | Self-disclosure | Likert scale | Self | 222 students in the US | Cross-sectional surveys | Individual | Individual-level | • Extroversion<br>• Perceived Internet risks<br>• Perceived critical mass<br>• Privacy value<br>• Attitude | None | SNS | Social media app | *Decision Support Systems* |
| 76 | Privacy disclosure | Likert scale | Self | 171 participants in China | Cross-sectional surveys | Individual | Individual-level | • Information sensitivity<br>• Privacy risks<br>• Information control<br>• Subjective norms | Theory of planned behavior<br>Privacy calculus theory | SNS | Social media app | *Electronic Commerce Research* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 77 | Actual location information disclosure | Coding | Self | 143 smartphone users | Experiment | Individual | Individual-level | • Privacy concerns<br>• Privacy benefits<br>• Information delivery mechanism<br>• Social proof<br>• Privacy concerns<br>• Trusting beliefs | None | SNS | Location-based service technology | *Electronic Markets* |
| 78 | Willingness to share personal data | Likert scale | Self | 3107 participants | Cross-sectional surveys | Individual | Individual-level | • Monetary compensation<br>• Data types<br>• Contexts | None | E-commerce | None | *Electronic Markets* |
| 79 | Self-disclosure | Likert scale | Self | 395 participants | Cross-sectional surveys | Individual | Individual-level | • Cognitive social identity<br>• Affective social identity<br>• Evaluative social identity<br>• Switching cost<br>• Dependency<br>• Trust | Social exchange theory<br>Social influence theory<br>Privacy calculus theory<br>Social identity theory | SNS | Social media app | *European Journal of Information Systems* |
| 80 | Self-disclosure | Likert scale | Self | 529 participants from the UK and France | Cross-sectional surveys | Individual | Individual-level | • Social influence<br>• Perceived reciprocity<br>• Perceived online community trust<br>• Privacy risk beliefs<br>• Perceived anonymity<br>• Perceived collectivism<br>• Perceived individualism | Social penetration theory<br>Social exchange theory<br>Hofstede's cultural dimensions | SNS | Social media app | *European Journal of Information Systems* |
| 81 | Self-disclosure | Likert scale | Self | 822 participants | Cross-sectional surveys | Individual | Individual-level | • Extroversion<br>• Perceived network assistance<br>• perceived cyber risk<br>• Social influence | None | SNS | Social media app | *European Journal of Information Systems* |
| 82 | Intention to disclose private information | Likert scale | Self | 667 participants | Cross-sectional surveys | Individual | Individual-level | • Adequacy<br>• Availability of information<br>• Website information quality<br>• Design appeal<br>• Reputation<br>• Privacy concerns<br>• Trust | Elaboration Likelihood Model | E-commerce | Website | *European Journal of Information Systems* |
| 83 | Information disclosure | Likert scale | Self | 314 Facebook users in the US | Cross-sectional surveys | Individual | Individual-level | • Privacy experiences<br>• Privacy awareness<br>• Privacy concerns<br>• Risks<br>• Trust<br>• benefits | Privacy calculus theory | SNS | Social media app | *European Journal of Information Systems* |
| 84 | Self-disclosure | Likert scale | Self | 547 participants in Rwanda | Cross-sectional surveys | Individual | Individual-level | • Privacy concerns<br>• Privacy risks<br>• Privacy control<br>• Trusting beliefs<br>• Perceived effectiveness of privacy policies<br>• Perceived effectiveness of organizational self-regulation | None | The Internet in general | Social networking sites, E-commerce website, E-government website, and E-health website | *Government Information Quarterly* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 85 | Private information disclosure | Likert scale | Self | 230 car drivers | Cross-sectional surveys | Individual | Individual-level | • Gender<br>• Age<br>• Locatability<br>• Previous privacy experience<br>• Perceived benefits<br>• Privacy concerns<br>• Perceived effectiveness of privacy policies | Communication privacy management theory | AI | Intelligent voice assistant | *Industrial Management & Data Systems* |
| 86 | Information disclosure | Likert scale | Self | 251 Chinese university students | Cross-sectional surveys | Individual | Individual-level | • Referent network size<br>• Perceived<br>• Complementarity<br>• Perceived external<br>• Prestige<br>• Perceived usefulness<br>• Habit<br>• Subjective norms | Technology acceptance model | SNS | Social media app | *Industrial Management & Data Systems* |
| 87 | Disclosure of personal information | Likert scale | Self | 547 participants | Interviews Cross-sectional surveys | Individual | Individual-level | • Privacy awareness<br>• Previous online invasion<br>• Mobile payment security<br>• Negative media exposure<br>• Perceived risks<br>• Perceived benefits<br>• Immediate gratification | Privacy calculus theory | Sharing economy | IT-enabled ride sharing | *Information & Management* |
| 88 | Personal health information disclosure intention | Likert scale | Self | 337 participants in China | Cross-sectional surveys | Individual | Individual-level | • Perceived severity<br>• Perceived vulnerability<br>• Response efficacy<br>• Self-efficacy<br>• Information support<br>• Emotion support<br>• Health information privacy concerns<br>• Perceived health status | Privacy calculus theory<br>Protection motivation theory | E-health | Website | *Information & Management* |
| 89 | Exposing others' information | Likert scale | Others | 1121 participants from the US and Korea | Cross-sectional surveys | Individual | Individual-level | • Individualistic/ collectivistic cultural orientation<br>• Individual privacy concerns<br>• Self-efficacy<br>• Severity of exposing others<br>• Susceptibility of exposing others | Protection motivation theory<br>Hofstede's cultural dimensions | SNS | Social media app | *Information & Management* |
| 90 | Intention to disclose information | Likert scale | Self | 367 students in the US | Experiment | Individual | Individual-level | • Previous online privacy invasion<br>• Internet privacy concerns<br>• Trust in website<br>• Big Five personality<br>• Context | Prospect theory<br>Theory of reasoned action | E-commerce | Website | *Information & Management* |
| 91 | Information disclosure | Likert scale | Self | 529 WeChat users in China | Cross-sectional surveys | Individual | Individual-level | • Privacy benefits: self-expression, life documentation, social rewards<br>• Privacy risks<br>• Dual motivation systems | Dual process model<br>Privacy calculus theory | SNS | Social media app | *Information & Management* |
| 92 | Information sharing behaviour | Likert scale | Self | 568 participants | Longitudinal surveys | Individual | Individual-level | • Perceived member support<br>• Enjoyment | Social exchange theory | SNS | Social media app | *Information & Management* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | • Informativeness<br>• Self-efficacy<br>• Community receptivity<br>• Relationship duration<br>• Norms of reciprocity | | | | |
| 93 | Information sharing behaviour | Likert scale | Self | 140 MBA students in China | Cross-sectional surveys | Individual | Individual-level | • Attitude toward Guanxi<br>• Collectivism<br>• Confusion dynamism | Interdependence theory<br>Social network theory<br>Hofstede's cultural dimension | The Internet in general | None | *Information & Management* |
| 94 | Intention to disclose information | Likert scale | Self | 264 students in the US | Experiment | Individual | Individual-level | • Trust<br>• Risk<br>• Relevance<br>• Attitude<br>• Usefulness | Theory of reasoned action<br>Transaction cost theory | E-commerce | Website | *Information & Management* |
| 95 | Self-disclosure | Likert scale | Self | 441 microblogging users in China | Cross-sectional surveys | Individual | Individual-level | • Perceived privacy risks<br>• Trust in the service provider<br>• Perceived anonymity self<br>• Convenience of relationship maintenance<br>• Relationship building<br>• Enjoyment<br>• Self-presentation | Social exchange theory<br>Privacy calculus theory | SNS | Microblog | *Information & Management* |
| 96 | Intention to share information | Likert scale | Self | 1089 participants in China | Cross-sectional surveys | Individual | Individual-level | • Anticipated extrinsic rewards<br>• Anticipated reciprocal relationship<br>• Knowledge self-efficacy<br>• Enjoyment in helping others<br>• Community climate: equity | Motivation theory<br>Social exchange theory<br>Equity theory | E-commerce | Website | *Information Systems Frontiers* |
| 97 | Self-disclosure | Likert scale | Self | 222 undergraduate students in the US | Cross-sectional surveys | Individual | Individual-level | • Trust<br>• Reciprocity<br>• Identification | Social capital theory | SNS | Social media app | *Information Systems Frontiers* |
| 98 | Information disclosure | Checklist | Self | 889 students in the US | Experiment | Individual | Individual-level | • Self-efficacy<br>• Perceived risks<br>• Perceived benefits<br>• Privacy concerns<br>• Privacy settings | Social cognitive theory<br>Privacy calculus theory | E-commerce | Mobile app | *Information Systems Journal* |
| 99 | Self-disclosure | Coding | Self | 458 students in China | Experiment | Individual | Individual-level | • Habit<br>• Role conflict<br>• Emotion<br>• Privacy risks<br>• Privacy control | Communication privacy management theory<br>Dual process model<br>Social role theory | SNS | Social media app | *Information Systems Journal* |
| 100 | Information disclosure | Coding | Self | 563 participants | Experiment | Individual | Individual-level | • Conversational style<br>• Reciprocation trigger<br>• Social presence<br>• Privacy concerns | Social presence theory | AI | Chatbot | *Information Systems Journal* |
| 101 | Privacy disclosure | Coding | Self | 1260 participants | Experiment | Individual | Individual-level | • The context of information disclosure<br>• Secondary use of the information<br>• Requirements to disclose information | None | The Internet in general | None | *Information Systems Research* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 102 | Willingness to disclose personal health information | Likert scale | Self | 1089 participants | Experiment | Individual | Individual-level | • Type of information<br>• Intended purpose<br>• Requesting stakeholder<br>• Health information privacy concerns<br>• Trust<br>• Health status emotions | Communication privacy management theory<br>Privacy calculus theory | E-health | Digital databanks | *Information Systems Research* |
| 103 | Willingness to disclose information | Likert scale | Self | 145 participants | Experiment | Individual | Individual-level | • Intrusiveness<br>• Attractiveness of benefits<br>• Timing | Privacy calculus theory | The Internet in general | None | *Information Systems Research* |
| 104 | Disclosure behaviors | Checklist | Self | 581 participants in the US | Experiment | Individual | Individual-level | • Privacy concerns<br>• Mood states<br>• Cognitive resources | Elaboration likelihood model | The Internet in general | None | *Information Systems Research* |
| 105 | Willingness to provide information | Likert scale | Self | 369 participants | Cross-sectional surveys | Individual | Individual-level | • Perceived Internet privacy risks<br>• Internet privacy concerns<br>• Internet trust<br>• Personal Internet interest | Privacy calculus theory | The Internet in general | Website | *Information Systems Research* |
| 106 | Information disclosure | Checklist | Self | 372 participants in China | Experiment | Individual | Individual-level | • Trust propensity<br>• Reputation<br>• Privacy experience | Justice theory | The Internet in general | Mobile app | *Information Systems Research* |
| 107 | Self-disclosure | Likert scale | Self | 251 participants | Cross-sectional surveys | Individual | Individual-level | • Perceived anonymity self<br>• Perceived anonymity of others<br>• Perceived media richness<br>• Perceived intrusiveness<br>• Privacy concerns<br>• Social rewards | Privacy calculus theory | SNS | Social media app | *Information Systems Research* |
| 108 | Information disclosure | Checklist | Self | 109 students in Singapore | Experiment | Individual | Individual-level | • Privacy statement<br>• Privacy seal<br>• Information sensitivity<br>• Monetary incentives<br>• Trust propensity<br>• Privacy concerns<br>• Prior experience | Contemporary choice theory | E-commerce | Website | *MIS Quarterly* |
| 109 | Private information disclosure | Checklist | Self | 412 participants | Experiment | Individual | Individual-level | • Privacy concerns<br>• Social perception<br>• Social cues<br>• Trust | Privacy calculus theory | SNS | Social media app | *MIS Quarterly* |
| 110 | Sharing of personal driving data | Coding | Self | 777 participants | Interviews<br>Cross-sectional surveys<br>Experiment | Individual | Individual-level | • Psychological ownership<br>• Relational trust<br>• Data sensitivity<br>• Data security<br>• Privacy concerns<br>• Self-efficacy<br>• Self-image congruency | Privacy calculus theory | IoT | Connected car | *MIS Quarterly* |
| 111 | Willingness to be profiled | Likert scale | Self | 523 participants | Cross-sectional surveys | Individual | Individual-level | • Demographics<br>• Previous online invasion<br>• Privacy concerns<br>• Importance of privacy policy<br>• Importance of information transparency | None | E-commerce | Website | *MIS Quarterly* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 112 | Self-disclosure | Likert scale | Self | 726 packages of parent-child data in China | Cross-sectional surveys | dyad | Individual-level | • Parental online privacy concerns<br>• Adolescents' online privacy concerns<br>• Internet evaluative mediation<br>• Gender of adolescent<br>• Parent-child dissonance | Privacy calculus theory | SNS | Social media app | *MIS Quarterly* |
| 113 | Self-disclosure | Likert scale | Self | 259 participants | Interviews<br>Cross-sectional surveys | Individual | Individual-level | • Trust in OSN members<br>• Perceived control<br>• Trust in OSN providers<br>• Perceived privacy risks<br>• Convenience<br>• Relationship building<br>• Self-presentation<br>• Enjoyment | Privacy calculus theory | SNS | Social media app | *Journal of Information Technology* |
| 114 | Privacy disclosure | Likert scale | Self | 500 Douyin adolescent users | Cross-sectional surveys | Individual | Individual-level | • Parental mediation<br>• Perceived privacy risks | Communication privacy management theory | SNS | Social media app | *Internet Research* |
| 115 | Self-disclosure | Likert scale | Self | 405 participants | Cross-sectional surveys | Individual | Individual-level | • Trust in OSN members<br>• Perceived control<br>• Trust in OSN providers<br>• Perceived privacy risks<br>• Convenience<br>• Relationship building<br>• Self-presentation<br>• Enjoyment<br>• Social influence | Social exchange theory<br>Privacy calculus theory | SNS | Social media app | *Internet Research* |
| 116 | Intention to share others' information | Likert scale | Others | 326 participants | Experiment | Individual | Individual-level | • Envy | None | SNS | Social media app | *Internet Research* |
| 117 | Self-disclosure | Likert scale | Self | 207 students in China | Cross-sectional surveys | Individual | Individual-level | • Centrality<br>• Trust<br>• Reciprocity<br>• Social norm<br>• Shared vision<br>• Big Five personality | Social capital theory | SNS | Social media app | *Internet Research* |
| 118 | Willingness to disclose health information | Likert scale | Self | 493 participants | Cross-sectional surveys | Individual | Individual-level | • Perceived transparency of privacy policies<br>• Cognitive trust<br>• Emotional trust | Theory of reasoned action | E-health | HIE technology | *Internet Research* |
| 119 | Self-disclosure | Likert scale | Self | 188 participants | Longitudinal surveys | Individual | Individual-level | • Privacy concerns<br>• Privacy risks<br>• Laziness | None | SNS | Social media app | *Internet Research* |
| 120 | Information disclosure | Likert scale | Self | 309 participants | Cross-sectional surveys | Individual | Individual-level | • Data breach victimization<br>• Peer privacy concerns<br>• Institutional privacy concerns<br>• Privacy risks<br>• Benefits<br>• Trust | Privacy calculus theory | The Internet in general | None | *Information Technology & People* |
| 121 | Disclosure about others | Likert scale | Others | 216 participants | Cross-sectional surveys | Individual | Individual-level | • Trust<br>• Privacy perceptions<br>• Benefits | Privacy calculus theory | SNS | Social media app | *Information Technology & People* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 122 | Willingness to disclose personal information | Likert scale | Self | 1563 participants in the US | Cross-sectional surveys | Individual | Individual-level | • Improper access<br>• Perceived effectiveness of privacy policies<br>• Perceived effectiveness of industry self-regulation<br>• Perceived privacy concerns<br>• Perceived security<br>• Perceived usefulness<br>• Perceived ease of use | Privacy calculus theory | Cloud computing | Mobile app | *Information Technology & People* |
| 123 | Self-disclosure | Likert scale | Self | 473 participants in the US and China | Cross-sectional surveys | Individual | Individual-level | • Trust in SNS provider<br>• Trust in SNS user<br>• Social rewards<br>• Privacy risks | Privacy calculus theory<br>Hofstede's cultural dimension | SNS | Social media app | *Information Technology & People* |
| 124 | Information disclosure | Likert scale | Self | 514 participants | Cross-sectional surveys | Individual | Individual-level | • Control over personal information<br>• Use of information<br>• Privacy notice | Theory of reasoned action<br>Theory of planned behavior | SNS | Social media app | *Information Technology & People* |
| 125 | Self-disclosure | Likert scale | Self | 54 participants | Cross-sectional surveys | Individual | Individual-level | • Perceived publicness | None | SNS | Social media app | *Information Technology & People* |
| 126 | Self-disclosure | Likert scale | Self | 378 participants | Cross-sectional surveys | Individual | Cross-level | • Trust<br>• Norm of reciprocity<br>• Identification<br>• Need to be on trend<br>• Need for popularity | Social capital theory | SNS | Online community | *Information Technology & People* |
| 127 | Self-disclosure | Likert scale | Self | 558 participants | Cross-sectional surveys | Individual | Individual-level | • Emotional support<br>• Information support<br>• Esteem support<br>• Network support<br>• Instrumental support | Social exchange theory | SNS | Social media app | *Information Technology & People* |
| 128 | Self-disclosure | Likert scale | Self | 2210 participants in Indonesia | Cross-sectional surveys | Individual | Individual-level | • Use of information<br>• Personal innovativeness<br>• Trust<br>• Privacy notice<br>• Privacy control<br>• Interactivity<br>• Perceived benefits<br>• Perceived risks | Communication privacy management theory<br>Privacy calculus theory | SNS | Social media app | *Information Technology & People* |
| 129 | Self-disclosure | Likert scale | Self | 213 participants | Experiment | Individual | Individual-level | • Information accessibility<br>• Social capital bridging<br>• Social capital bonding<br>• Perceived privacy risks | Social exchange theory<br>Privacy calculus theory | SNS | Social media app | *Information Technology & People* |
| 130 | Self-disclosure | Likert scale | Self | 647 participants in China | Cross-sectional surveys | Individual | Individual-level | • Privacy concerns<br>• Perceived effectiveness of privacy settings<br>• Perceived effectiveness of privacy policies<br>• Perceived effectiveness of industry self-regulation<br>• Perceived effectiveness of government legislation<br>• Network externality<br>• Technology complementarity | Control agency theory<br>Network effect theory | E-commerce | Mobile payment technology | *Information Technology & People* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 131 | Willingness to share personal data | Likert scale | Self | 6464 participants in the US, China, and Germany | Cross-sectional surveys | Individual | Individual-level | • Health concerns<br>• Psychological effect<br>• Disclosure benefits<br>• Disclosure risks<br>• Responsibility<br>• Trust in government | None | E-health | Contact tracing technology | *Information, Communication & Society* |
| 132 | Willingness to disclose personal information | Likert scale | Self | 1522 participants | Experiment<br>Cross-sectional surveys | Individual | Individual-level | • Subjective norm<br>• Image<br>• Output quality<br>• Result demonstrability<br>• Perceived prosocial benefit<br>• Trust in public institution<br>• Fear of covid-19<br>• Perceived ease of use | Technology acceptance model | E-health | Contact tracing technology | *Information, Communication & Society* |
| 133 | Willingness to provide information | Likert scale | Self | 178 participants | Experiment | Individual | Individual-level | • Trust<br>• Monetary compensation<br>• Information sensitivity | None | E-commerce | None | *International Journal of Electronic Commerce* |
| 134 | Information sharing behavior | Likert scale | Self | 405 participants | Cross-sectional surveys | Individual | Individual-level | • Interactivity<br>• Social presence<br>• Commitment<br>• Privacy risks<br>• Subjective norms<br>• Social ties | Theory of reasoned action<br>Social capital theory | SNS | Social media app | *International Journal of Electronic Commerce* |
| 135 | Intention to disclose location-based information | Likert scale | Self | 368 participants in China | Cross-sectional surveys | Individual | Individual-level | • Incentive provision<br>• Interaction promotion<br>• Privacy control<br>• Privacy policy<br>• Privacy concerns<br>• Big Five personality<br>• Privacy benefits<br>• Personal innovativeness | Privacy calculus theory<br>Justice theory | SNS | Location-based service technology | *International Journal of Electronic Commerce* |
| 136 | Willingness to share information | Likert scale | Self | 685 participants in the UK | Cross-sectional surveys | Individual | Individual-level | • Disposition of privacy<br>• Privacy awareness<br>• Perceived privacy control<br>• Trust<br>• Privacy experience<br>• Privacy knowledge<br>• Privacy protection regulation<br>• Privacy concerns<br>• Personalization<br>• Sensitivity of data | Privacy calculus theory | The Internet in general | Traveling service technology | *International Journal of Information Management* |
| 137 | Intention to share information | Likert scale | Self | 405 university student participants in the US | Cross-sectional surveys | Individual | Individual-level | • Social presence<br>• Privacy risks<br>• Social ties<br>• Commitment<br>• Attitude toward information sharing<br>• Subjective norms<br>• Gender | Theory of reasoned action<br>Social role theory | SNS | Social media app | *International Journal of Information Management* |
| 138 | Intention to disclose personal information | Likert scale | Self | 327 participants | Cross-sectional surveys | Individual | Individual-level | • Perceived severity<br>• Perceived control<br>• Self-presentation<br>• Personalized services | Privacy calculus theory | The Internet in general | Mobile app | *International Journal of Information Management* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 139 | Self-disclosure | Likert scale | Self | 913 participants in China | Cross-sectional surveys | Individual | Individual-level | • Perceived benefits<br>• Perceived risks<br>• Application compatibility<br>• Application reputation<br>• Self-esteem<br>• Flow experience<br>• Monetary rewards<br>• Social rewards<br>• Privacy concerns | Privacy calculus theory | SNS | Social media app | *International Journal of Information Management* |
| 140 | Information sharing behavior | Likert scale | Self | 1177 participants in China | Cross-sectional surveys<br>Web scraping | Individual | Individual-level | • Reputation<br>• Enjoyment<br>• Structural capital<br>• Cognitive capital<br>• Relational capital | Social capital theory | E-commerce | Mobile app | *International Journal of Information Management* |
| 141 | Intention to disclose personal information | Likert scale | Self | 360 participants in China | Cross-sectional surveys | Individual | Individual-level | • Perceived benefits<br>• Perceived risks<br>• Perceived effectiveness of privacy settings<br>• Perceived effectiveness of privacy policies<br>• Perceived value<br>• Psychological comfort | Control agency theory<br>Privacy calculus theory | E-commerce | Mobile payment technology | *International Journal of Information Management* |
| 142 | Information disclosure | Likert scale | Self | 263 adolescent participants in Australia | Cross-sectional surveys | Individual | Individual-level | • Peer pressure<br>• Signaling<br>• Trust<br>• Privacy risks<br>• Design<br>• Relaxed attitude toward privacy | None | SNS | Social media app | *International Journal of Information Management* |
| 143 | Information disclosure intention | Likert scale | Self | 451 participants | Interviews<br>Cross-sectional surveys | Individual | Individual-level | • Structural assurance<br>• Surveillance<br>• Future risks<br>• Intrusion<br>• Property<br>• Physical<br>• Profiling<br>• Secondary use<br>• Social risks<br>• Stalking<br>• Permeability risks<br>• Ownership risks<br>• Linkage risks<br>• Privacy risks | Privacy calculus theory<br>Communication privacy management theory | SNS | Location-based service technology | *International Journal of Information Management* |
| 144 | Information disclosure intention | Likert scale | Self | 215 participants in China | Cross-sectional surveys | Individual | Individual-level | • Locatability<br>• Personalization<br>• Social benefits<br>• Gender | Self-determination theory<br>Privacy calculus theory<br>Social role theory | SNS | Location-based service technology | *International Journal of Information Management* |
| 145 | Communicating personal health information | Likert scale | Self | 235 participants in the US | Cross-sectional surveys | Individual | Individual-level | • Age<br>• Health status<br>• Affective commitment<br>• Privacy concerns<br>• Expected personal outcomes | Privacy calculus theory | E-health | PHI technology | *International Journal of Information Management* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | • Expected community outcomes | | | | |
| 146 | Self-disclosure | Likert scale | Self | 397 participants in South Korea | Cross-sectional surveys | Individual | Individual-level | • Algorithmic awareness<br>• Self-efficacy<br>• Privacy concerns<br>• Trust | Privacy calculus theory | AI | Algorithm-driven platforms | *International Journal of Information Management* |
| 147 | Willingness to share personal health information | Likert scale | Self | 170 older adults in South Korea | Cross-sectional surveys | Individual | Individual-level | • Demographics<br>• Health status<br>• Socio-economic information | None | E-health | None | *International Journal of Medical Informatics* |
| 148 | Willingness to share personal health information | Likert scale | Self | 220 pregnant women participant in South Korea | Cross-sectional surveys | Individual | Individual-level | • Demographics<br>• E-health literacy<br>• Socio-economic information<br>• Health status | None | E-health | None | *International Journal of Medical Informatics* |
| 149 | Willingness to share health data | Likert scale | Self | 386 participants | Cross-sectional surveys | Individual | Individual-level | • Demographics<br>• Health status<br>• Socio-economic information<br>• Ownership | None | E-health | None | *International Journal of Medical Informatics* |
| 150 | Willingness to disclose personal information | Likert scale | Self | 1858 participants | Cross-sectional surveys | Individual | Individual-level | • Privacy<br>• Trust<br>• Personalization value<br>• Customer benefits<br>• Type of data | Privacy calculus theory | E-commerce | Digital advertising | *Journal of Advertising Research* |
| 151 | Information disclosure | Likert scale | Self | 182 participants in the US | Experiment | Individual | Individual-level | • Perceived relevance<br>• Perceived usefulness<br>• Monetary rewards<br>• Privacy protection beliefs<br>• Privacy risks beliefs | Social contract theory<br>Privacy calculus theory | E-commerce | Website | *Journal of Computer Information Systems* |
| 152 | Personal information disclosure | Likert scale | Self | 186 participants in the US | Experiment | Individual | Individual-level | • Sales promotion<br>• Location congruence<br>• Perceived benefits<br>• Perceived costs | Privacy calculus theory | E-commerce | Location-based service technology | *Journal of Consumer Behavior* |
| 153 | Willingness to share private data | Likert scale | Self | 20,508 participants | Experiment | Individual | Individual-level | • Privacy concerns<br>• Type of data<br>• Data collector<br>• Type of reward<br>• Control and data anonymity | None | E-commerce | None | *Journal of Consumer Behavior* |
| 154 | Self-disclosure | Likert scale | Self | 851 participants | Cross-sectional surveys | Individual | Individual-level | • Self-brand connection<br>• Need to belong | None | E-commerce | None | *Journal of Consumer Behavior* |
| 155 | Willingness to share data about others | Likert scale | Others | 706 participants | Experiment | Individual | Individual-level | • Social value orientation<br>• Self-construal<br>• Psychological ownership | Self-construal theory | E-commerce | None | *Journal of Consumer Psychology* |
| 156 | Self-disclosure | Coding | Self | 1184 participants | Experiment | Individual | Individual-level | • Dyadic dissimilarity<br>• Relational costs<br>• Minority costs | None | SNS | Social media app | *Journal of Consumer Psychology* |
| 157 | Intention to disclose personal information | Likert scale | Self | 426 participants in China | Cross-sectional surveys | Individual | Individual-level | • Big Five personality<br>• Privacy fatigue<br>• Privacy concerns | None | The Internet in general | Mobile app | *Journal of Enterprise Information Management* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 158 | Self-disclosure | Likert scale | Self | 596 participants | Cross-sectional surveys | Individual | Individual-level | • Intrinsic gratification<br>• Social gratification<br>• Extrinsic gratification<br>• Gender | Use and gratification theory | The Internet in general | Mobile instant messaging | *Journal of Enterprise Information Management* |
| 159 | Intention to share information | Likert scale | Self | 412 participants | Cross-sectional surveys | Individual | Individual-level | • Personal information collection and usage<br>• Perceived effectiveness of privacy policies<br>• Personal information control<br>• Gender | None | SNS | Social media app | *Journal of Global Information Management* |
| 160 | Information sharing behavior | Likert scale | Self | 436 participants in China | Cross-sectional surveys | Individual | Individual-level | • Past sharing experience<br>• Trust<br>• Perceived benefits<br>• Perceived richness | Social cognitive theory<br>Social network theory | SNS | Social media app | *Journal of Global Information Management* |
| 161 | Self-disclosure | Likert scale | Self | 517 participants | Experiment | Individual | Individual-level | • Affect toward self-disclosure<br>• Affect toward SNS<br>• Self-presentation<br>• Expression<br>• Social acceptance<br>• Reciprocity<br>• Social rejection<br>• Privacy risks | Affect heuristics theory<br>Privacy calculus theory | SNS | Social media app | *Journal of Management Information Systems* |
| 162 | Self-disclosure | Likert scale | Self | 195 participants in Canada | Experiment | Individual | Individual-level | • Perceived transparency<br>• Perceived responsiveness<br>• Perceived outcome expectancy<br>• Perceived loss of privacy<br>• Trust<br>• Perceived interdependence<br>• Design features | Privacy calculus theory | AI | Virtual advisor chatbot | *Journal of Management Information Systems* |
| 163 | Intention to disclose information | Likert scale | Self | 286 participants in Germany | Experiment | Individual | Individual-level | • Disposition to value privacy<br>• Experience with the Internet<br>• Experience with online personalization | None | E-commerce | Website | *Journal of Management Information Systems* |
| 164 | Intention to disclose personal information | Likert scale | Self | 528 participants in Singapore | Experiment | Individual | Individual-level | • Compensation<br>• Industry self-regulation<br>• Government regulation<br>• Privacy benefits<br>• Privacy risks | Privacy calculus theory<br>Justice theory | E-commerce | Location-based service technology | *Journal of Management Information Systems* |
| 165 | Private disclosure | Likert scale | Self | 168 participants | Experiment | Individual | Individual-level | • Information privacy concerns<br>• Information trusting beliefs<br>• Information risks beliefs<br>• Information sensitivity | Communication privacy management theory | SNS | Social media app | *Journal of the Association for Information Systems* |
| 166 | Willingness to communicate personal health information | Likert scale | Self | 235 participants | Cross-sectional surveys | Individual | Individual-level | • Expected positive personal outcomes<br>• Expected positive community outcomes<br>• Privacy concerns<br>• Affective commitment | Privacy calculus theory | E-health | Online health community | *Journal of the Association for Information Systems* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 167 | Willingness to disclose personal information | Likert scale | Self | 1116 participants | Cross-sectional surveys | Individual | Individual-level | • Privacy policy permissiveness<br>• Privacy risks | None | SNS | Social media app | *Journal of Strategic Information Systems* |
| 168 | Intentions to disclose information | Likert scale | Self | 301 participants | Cross-sectional surveys | Individual | Individual-level | • Affect<br>• Internet security<br>• Website trust<br>• Website privacy | None | The Internet in general | None | *Journal of Strategic Information Systems* |
| 169 | Willingness to provide personal information | Likert scale | Self | 422 participants | Cross-sectional surveys | Individual | Individual-level | • Internet privacy concerns<br>• Perceived need for government surveillance<br>• Government intrusion concerns | None | The Internet in general | None | *Journal of Strategic Information Systems* |
| 170 | Willingness to give private information | Likert scale | Self | 140 participants in the US | Experiment | Individual | Individual-level | • Privacy statement<br>• Privacy seals<br>• Security seals<br>• Security features<br>• Trust | None | E-commerce | Website | *Journal of Strategic Information Systems* |
| 171 | Self-disclosure | Coding | Self | 270 student participants in the US | Experiment | Individual | Individual-level | • Implicit cues<br>• Explicit cues<br>• Privacy settings<br>• Information sensitivity | None | SNS | Social media app | *Journal of Computer-mediated Communication* |
| 172 | Self-disclosure | Likert scale | Self | 1131 participants in Dutch | Experiment | Individual | Individual-level | • Benefits<br>• Trust<br>• Risk perceptions<br>• Risk beliefs<br>• Privacy concerns | Privacy calculus theory | The Internet in general | Website | *Journal of Computer-mediated Communication* |
| 173 | Self-disclosure | Likert scale | Self | 2739 participants in Germany | Cross-sectional surveys | Individual | Individual-level | • Privacy concerns<br>• Social relevance<br>• Number of applications | None | SNS | Social media app | *Journal of Computer-mediated Communication* |
| 174 | Self-disclosure | Likert scale | Self | 1156 participants in the US | Cross-sectional surveys | Individual | Individual-level | • Facebook benefits<br>• Privacy concerns<br>• Self-efficacy | Privacy calculus theory | SNS | Social media app | *Journal of Computer-mediated Communication* |
| 175 | Self-disclosure | Likert scale | Self | 560 participants | Cross-sectional surveys | Individual | Individual-level | • Privacy concerns<br>• Stress<br>• Gender | Communication privacy management theory | SNS | Social media app | *Journal of Computer-mediated Communication* |
| 176 | Self-disclosure | Likert scale | Self | 742 Israeli Facebook users | Cross-sectional surveys | Individual | Individual-level | • Observed gains<br>• Observed losses<br>• Assessment of gains to Self<br>• Assessment of losses to Self | Observational learning theory | SNS | Social media app | *Journal of Computer-mediated Communication* |
| 177 | Information disclosure | Likert scale | Self | 622 teen-parent pairs | Cross-sectional surveys | Dyad | Individual-level | • Teen costs<br>• Teen benefits<br>• Teen interpersonal trust<br>• Parental mediation<br>• Parental concerns<br>• Parental interpersonal trust | Protection motivation theory<br>Privacy calculus theory<br>Social capital theory | SNS | Social media app | *Journal of the Association for Information Science and Technology* |
| 178 | Intention to give private information | Likert scale | Self | 362 participants in South Korea | Cross-sectional surveys | Individual | Individual-level | • Relationship management<br>• Self-presentation<br>• Privacy concerns<br>• Subjective norms | Privacy calculus theory<br>Social influence theory | SNS | Social media app | *Journal of the Association for Information Science and Technology* |
| 179 | Intention to give private information | Likert scale | Self | 396 participants in South Korea | Cross-sectional surveys | Individual | Individual-level | • Perceived information risks<br>• Perceived social risks<br>• Information misuse concerns | Communication privacy management theory | SNS | Social media app | *Journal of the Association for Information Science and Technology* |

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**Table D1** (*continued*)

| | Terminology | Measurement | Ownership | Samples | Method | Unit of data collection | Level of data analysis | Antecedents | Theories | Context | IT Artifact | Journal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | • Information misunderstanding concerns<br>• Usage period<br>• Proportion of offline friends | | | | |
| 180 | Self-disclosure | Likert scale | Self | 249 participants | Cross-sectional surveys | Individual | Individual-level | • Need for self-identity<br>• Privacy management<br>• Perceived privacy control | Social identity theory | SNS | Social media app | *Journal of the Association for Information Science and Technology* |
| 181 | Willingness to disclose personal information | Likert scale | Self | 1114 participants | Cross-sectional surveys | Individual | Individual-level | • Perceived surveillance<br>• Perceived control<br>• Propensity to trust<br>• Social influence<br>• Health benefits<br>• Perceived privacy<br>• Reciprocal benefits | Social influence theory<br>Privacy calculus theory | E-health | Contact tracing app | *Journal of the Association for Information Science and Technology* |
| 182 | Information disclosure | Checklist | Self | 390 participants | Experiment | Individual | Individual-level | • Privacy concerns<br>• Self-efficacy<br>• Satisfaction<br>• Perceived privacy protection | Elaboration likelihood model | The Internet in general | Mobile app | *Journal of the Association for Information Science and Technology* |
| 183 | Willingness to disclose information | Likert scale | Self | 500 participants | Cross-sectional surveys | Individual | Individual-level | • Information sensitivity<br>• Platform security<br>• Perceived usefulness<br>• Privacy concerns<br>• Trust in platforms<br>• Satisfaction | Privacy calculus theory | E-health | Online health community | *Technological Forecasting and Social Change* |
| 184 | Co-owned information disclosure | Likert scale | Co-owned | 740 participants | Cross-sectional surveys | Individual | Individual-level | • Personal benefit<br>• Personal privacy risks<br>• Others' benefit<br>• Others' privacy risks<br>• Relationship benefit<br>• Relationship privacy risks<br>• Personal distributive fairness<br>• Others' distributive fairness | Privacy calculus theory<br>Justice theory | SNS | Social media app | *Internet Research* |
| 185 | Private information disclosure | Likert scale | Self | 220 participants | Cross-sectional surveys | Dyad | Individual-level | • Susceptibility to social influence<br>• Perceived immediate rewards<br>• Perceived long-term rewards<br>• Privacy concerns<br>• Privacy vulnerability<br>• Privacy self-efficacy<br>• Response efficacy<br>• Impulsiveness<br>• Coping responses | Protection motivation theory | SNS | Social media app | *Journal of the Association for Information Systems* |
| 186 | Personal health information sharing | Likert scale | Self | 513 participants | Cross-sectional surveys | Individual | Individual-level | • Technology control<br>• Legislative control<br>• Community norms<br>• Perceived benefits<br>• Personal privacy norms<br>• Perceived altruistic value | Contextual integrity framework | E-health | Online health community | Journal of the Association for Information Systems |

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**Mingxin Zhang:** Mingxin Zhang is a PhD student at the School of Computing and Information Systems in the University of Melbourne. He studied Library Science and Information Science at Sun Yat-sen University from 2015 to 2022 and received his bachelor's degree and master's degree. His research interests include information privacy, privacy calculus, and social media user behaviours. His papers appear in conference proceedings such as the Australasian Conference on Information Systems and the Pacific-Asia Conference on Information Systems.

**Reeva Lederman:** Reeva Lederman is a Professor at the School of Computing and Information Systems in the University of Melbourne and has been a faculty member since 2000. She leads the Information Systems research group in CIS and in a member of the CIS Executive. She is an active member of the international health information systems research community. In 2012 she was awarded the prestigious Stafford Beer Medal for IS Research and in 2014 she won the Melbourne School of Engineering annual teaching award for Teaching Excellence. She has been an investigator on large NHMRC grants from 2015 and extending to 2022 in the area of online systems for mental health.

**Ofir Turel:** Ofir Turel is a Professor in Information Systems Management within the School of Computing and Information Systems at The University of Melbourne. He is also a Scholar in Residence at the Decision Neuroscience Program, University of Southern California. His research interests include a broad range of behavioural, managerial, and biophysiological issues related to the use and management of computer information systems and technologies. He has published over 160 journal papers in leading journals. They include such information systems journals as MIS Quarterly, Journal of Management Information Systems, MIT Sloan Management Review, Journal of AIS, European Journal of Information Systems, Information Systems Journal, and Communications of the ACM. Example psychology and neuroscience outlets include Journal of Psychiatric Research, Addiction Biology, Cognitive, Affective & Behavioral Neuroscience, Behavioural Brain Research, and Social Neuroscience.